

Optimization of the activity and biodegradability of ionizable lipids for mRNA delivery via directed chemical evolution

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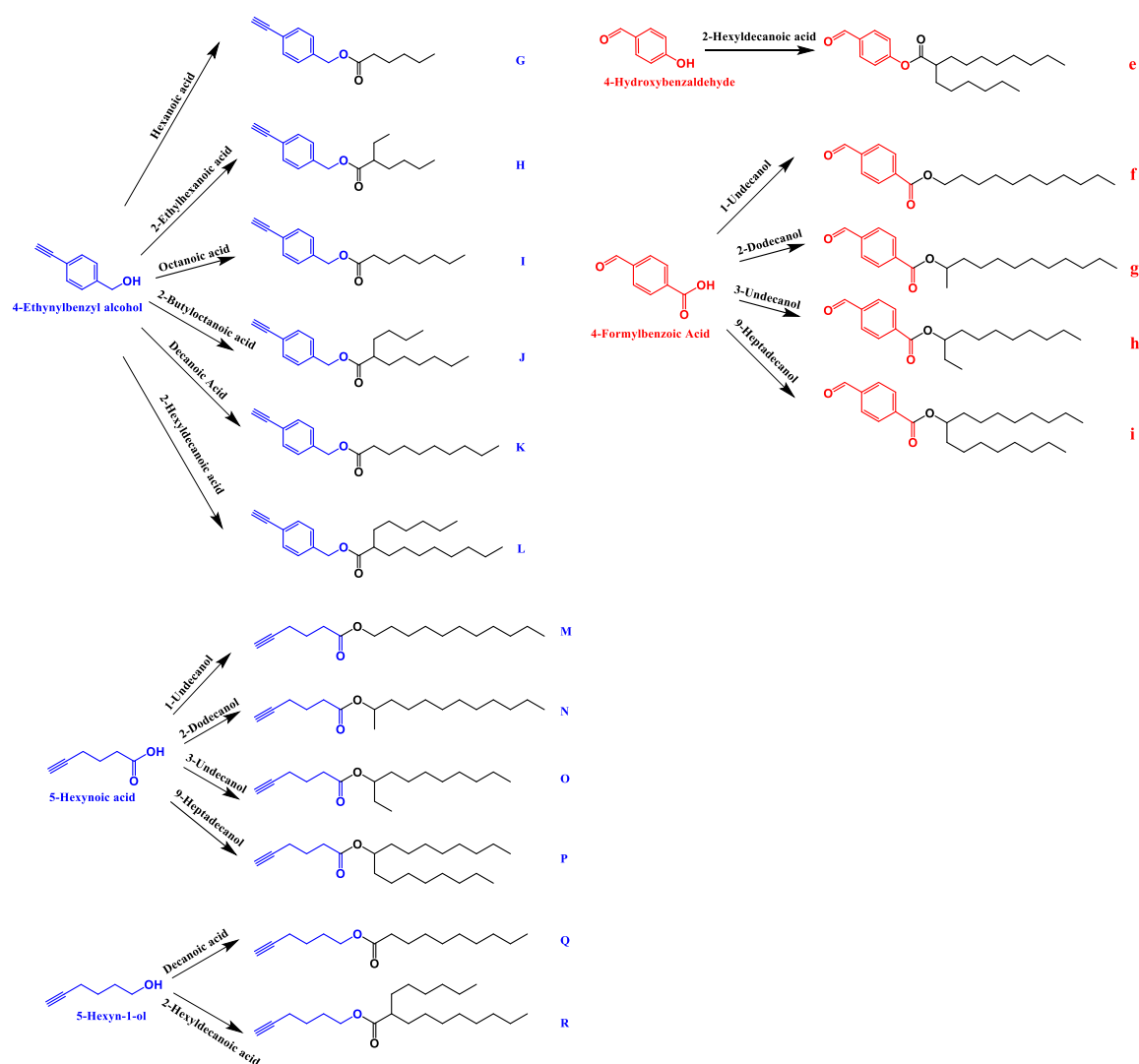
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Supplementary Method 1 | General synthesis of biodegradable aldehydes and alkynes. Biodegradable aldehyde or alkyne tail was synthesized via a one-step, DCC/DMAP-mediated esterification reaction. Briefly, alcohol (2 mmol, 1 eq.), acid (2.2 mmol, 1.1 eq.), DCC (2.2 mmol, 1.1 eq.) and DMAP (0.4 mmol, 0.2 eq.) were dissolved in 20 ml anhydrous DCM in a round bottom flask and stirred at RT under N₂ protection for 24 h. The reaction mixture was filtered and the filtrate was evaporated under vacuum. The residue was separated using a CombiFlash NextGen 300+ chromatography system with gradient elution from hexane to 20:80 ethyl acetate/hexane (aq) over 15 min to give desired products.

G: Colorless oil, yield 91%. MS-ESI: calculated for C₁₅H₁₈O₂: 230.13, found [M+H]⁺ = 231.12. ¹H NMR (400 MHz, CDCl₃) δ 7.51 – 7.42 (m, 2H), 7.31 – 7.27 (m, 2H), 5.11 (s, 2H), 3.07 (s, 1H), 2.34 (t, J = 7.5 Hz, 2H), 1.63 (p, J = 7.3 Hz, 2H), 1.35 – 1.21 (m, 4H), 0.89 (t, J = 6.8 Hz, 3H).

H: Colorless oil, yield 90%. MS-ESI: calculated for C₁₇H₂₂O₂: 258.16, found [M+H]⁺ = 259.14. ¹H NMR (400 MHz, CDCl₃) δ 7.50 – 7.43 (m, 2H), 7.34 – 7.28 (m, 2H), 5.10 (s, 2H), 3.09 (s, 1H), 2.37 (tt, J = 8.9, 5.3 Hz, 1H), 1.58 (dq, J = 12.7, 6.6 Hz, 2H), 1.44 (ddd, J = 14.2, 8.1, 5.2 Hz, 2H), 1.25 (h, J = 6.4 Hz, 4H), 0.88 (td, J = 6.9, 4.2 Hz, 6H).

I: Colorless oil, yield 92%. MS-ESI: calculated for C₁₇H₂₂O₂: 258.16, found [M+H]⁺ = 259.15. ¹H NMR (400 MHz, CDCl₃) δ 7.51 – 7.44 (m, 2H), 7.33 – 7.27 (m, 2H), 5.11 (s, 2H), 3.07 (s, 1H), 2.34 (t, J = 7.5 Hz, 2H), 1.65 (p, J = 7.3 Hz, 2H), 1.37 – 1.22 (m, 8H), 0.87 (t, J = 6.8 Hz, 3H).

J: Colorless oil, yield 91%. MS-ESI: calculated for C₂₁H₃₀O₂: 314.22, found [M+H]⁺ = 315.21. ¹H NMR (400 MHz, CDCl₃) δ 7.52 – 7.43 (m, 2H), 7.33 – 7.27 (m, 2H), 5.10 (s, 2H), 3.07 (s, 1H), 2.38 (tt, J = 8.9, 5.3 Hz, 1H), 1.60 (dq, J = 12.7, 6.6 Hz, 2H), 1.46 (ddd, J = 14.2, 8.1, 5.2 Hz, 2H), 1.24 (h, J = 6.4 Hz, 12H), 0.88 (td, J = 6.9, 4.2 Hz, 6H).

K: Colorless oil, yield 92%. MS-ESI: calculated for $C_{19}H_{26}O_2$: 286.19, found $[M+H]^+ = 287.17$. 1H NMR (400 MHz, $CDCl_3$) δ 7.52 – 7.44 (m, 2H), 7.33 – 7.28 (m, 2H), 5.10 (s, 2H), 3.08 (s, 1H), 2.35 (t, $J = 7.5$ Hz, 2H), 1.64 (p, $J = 7.3$ Hz, 2H), 1.36 – 1.20 (m, 12H), 0.88 (t, $J = 6.8$ Hz, 3H).

L: Brown oil, yield 90%. MS-ESI: calculated for $C_{25}H_{38}O_2$: 370.29, found $[M+H]^+ = 371.30$. 1H NMR (400 MHz, $CDCl_3$) δ 7.51 – 7.44 (m, 2H), 7.34 – 7.27 (m, 2H), 5.11 (s, 2H), 3.08 (s, 1H), 2.38 (tt, $J = 8.9, 5.3$ Hz, 1H), 1.59 (dq, $J = 12.7, 6.6$ Hz, 2H), 1.45 (ddd, $J = 14.2, 8.1, 5.2$ Hz, 2H), 1.24 (h, $J = 6.4$ Hz, 20H), 0.87 (td, $J = 6.9, 4.2$ Hz, 6H).

M: Colorless oil, yield 94%. MS-ESI: calculated for $C_{17}H_{30}O_2$: 266.22, found $[M+H]^+ = 267.21$. 1H NMR (400 MHz, $CDCl_3$) δ 4.10 (t, $J = 6.5$ Hz, 2H), 2.28 (t, $J = 7.5$ Hz, 2H), 2.25 (td, $J = 7.0, 2.7$ Hz, 2H), 1.97 (t, $J = 2.7$ Hz, 1H), 1.80 – 1.71 (m, 2H), 1.67 – 1.54 (m, 2H), 1.37 – 1.22 (m, 16H), 0.92 – 0.81 (m, 3H).

N: Colorless oil, yield 91%. MS-ESI: calculated for $C_{18}H_{32}O_2$: 280.24, found $[M+H]^+ = 281.25$. 1H NMR (400 MHz, $CDCl_3$) δ 4.84 (tt, $J = 6.9, 5.5$ Hz, 1H), 2.43 (t, $J = 7.4$ Hz, 2H), 2.28 (td, $J = 7.0, 2.6$ Hz, 2H), 1.97 (t, $J = 2.7$ Hz, 1H), 1.87 (p, $J = 7.2$ Hz, 2H), 1.61 – 1.49 (m, 2H), 1.26 (m, 19H), 0.91 – 0.82 (m, 3H).

O: Colorless oil, yield 91%. MS-ESI: calculated for $C_{17}H_{30}O_2$: 266.22, found $[M+H]^+ = 267.23$. 1H NMR (400 MHz, $CDCl_3$) δ 4.83 (tt, $J = 6.9, 5.5$ Hz, 1H), 2.44 (t, $J = 7.4$ Hz, 2H), 2.27 (td, $J = 7.0, 2.6$ Hz, 2H), 1.97 (t, $J = 2.7$ Hz, 1H), 1.86 (p, $J = 7.2$ Hz, 2H), 1.62 – 1.49 (m, 4H), 1.26 (m, 12H), 0.91 – 0.84 (m, 6H).

P: Brown oil, yield 90%. MS-ESI: calculated for $C_{23}H_{42}O_2$: 350.32, found $[M+H]^+ = 351.30$. 1H NMR (400 MHz, $CDCl_3$) δ 4.88 (p, $J = 6.3$ Hz, 1H), 2.43 (t, $J = 7.4$ Hz, 2H), 2.26 (td, $J = 7.0, 2.7$ Hz, 2H), 1.96 (t, $J = 2.7$ Hz, 1H), 1.85 (p, $J = 7.2$ Hz, 2H), 1.55 – 1.46 (m, 4H), 1.26 (s, 24H), 0.91 – 0.85 (m, 6H).

Q: Colorless oil, yield 92%. MS-ESI: calculated for $C_{16}H_{28}O_2$: 252.21, found $[M+H]^+ = 253.19$. 1H NMR (400 MHz, $CDCl_3$) δ 4.09 (t, $J = 6.5$ Hz, 2H), 2.29 (t, $J = 7.5$ Hz, 2H), 2.23 (td, $J = 7.0, 2.7$ Hz, 2H), 1.95 (t, $J = 2.7$ Hz, 1H), 1.80 – 1.70 (m, 2H), 1.66 – 1.54 (m, 4H), 1.36 – 1.23 (m, 12H), 0.88 (td, $J = 6.9, 1.8$ Hz, 3H).

R: Light yellow oil, yield 90%. MS-ESI: calculated for $C_{22}H_{40}O_2$: 336.30, found $[M+H]^+ = 337.30$. 1H NMR (400 MHz, $CDCl_3$) δ 4.10 (t, $J = 6.4$ Hz, 2H), 2.31 (tt, $J = 8.9, 5.3$ Hz, 1H), 2.23 (td, $J = 7.0, 2.6$ Hz, 2H), 1.95 (t, $J = 2.6$ Hz, 1H), 1.82 – 1.54 (m, 6H), 1.43 (dt, $J = 7.9, 5.6$ Hz, 2H), 1.26 (d, $J = 4.2$ Hz, 20H), 0.92 – 0.83 (m, 6H).

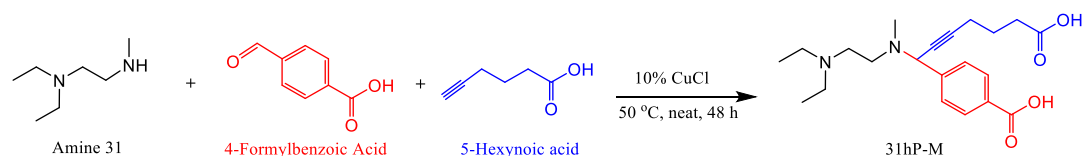
e: Colorless oil, yield 90%. MS-ESI: calculated for $C_{23}H_{36}O_3$: 360.27, found $[M+H]^+ = 361.25$. 1H NMR (400 MHz, $CDCl_3$) δ 10.00 (s, 1H), 7.97 – 7.88 (m, 2H), 7.27 – 7.22 (m, 2H), 2.60 (tt, $J = 8.9, 5.4$ Hz, 1H), 1.83 – 1.69 (m, 2H), 1.66 – 1.53 (m, 2H), 1.46 – 1.22 (m, 20H), 0.89 (td, $J = 6.9, 4.4$ Hz, 6H).

f: Colorless oil, yield 94%. MS-ESI: calculated for $C_{19}H_{28}O_3$: 304.20, found $[M+H]^+ = 305.22$. 1H NMR (400 MHz, $CDCl_3$) δ 10.11 (s, 1H), 8.23 – 8.18 (m, 2H), 8.01 – 7.92 (m, 2H), 4.10 (t, $J = 6.5$ Hz, 2H), 2.25 (td, $J = 7.0, 2.7$ Hz, 2H), 1.49 – 1.19 (m, 16H), 0.92 – 0.82 (m, 3H).

g: Colorless oil, yield 91%. MS-ESI: calculated for $C_{20}H_{30}O_3$: 318.22, found $[M+H]^+ = 319.23$. 1H NMR (400 MHz, $CDCl_3$) δ 10.12 (s, 1H), 8.23 – 8.18 (m, 2H), 8.01 – 7.92 (m, 2H), 5.12 (m, 1H), 1.81 – 1.60 (m, 2H), 1.48 – 1.18 (m, 19H), 0.91 – 0.83 (m, 3H).

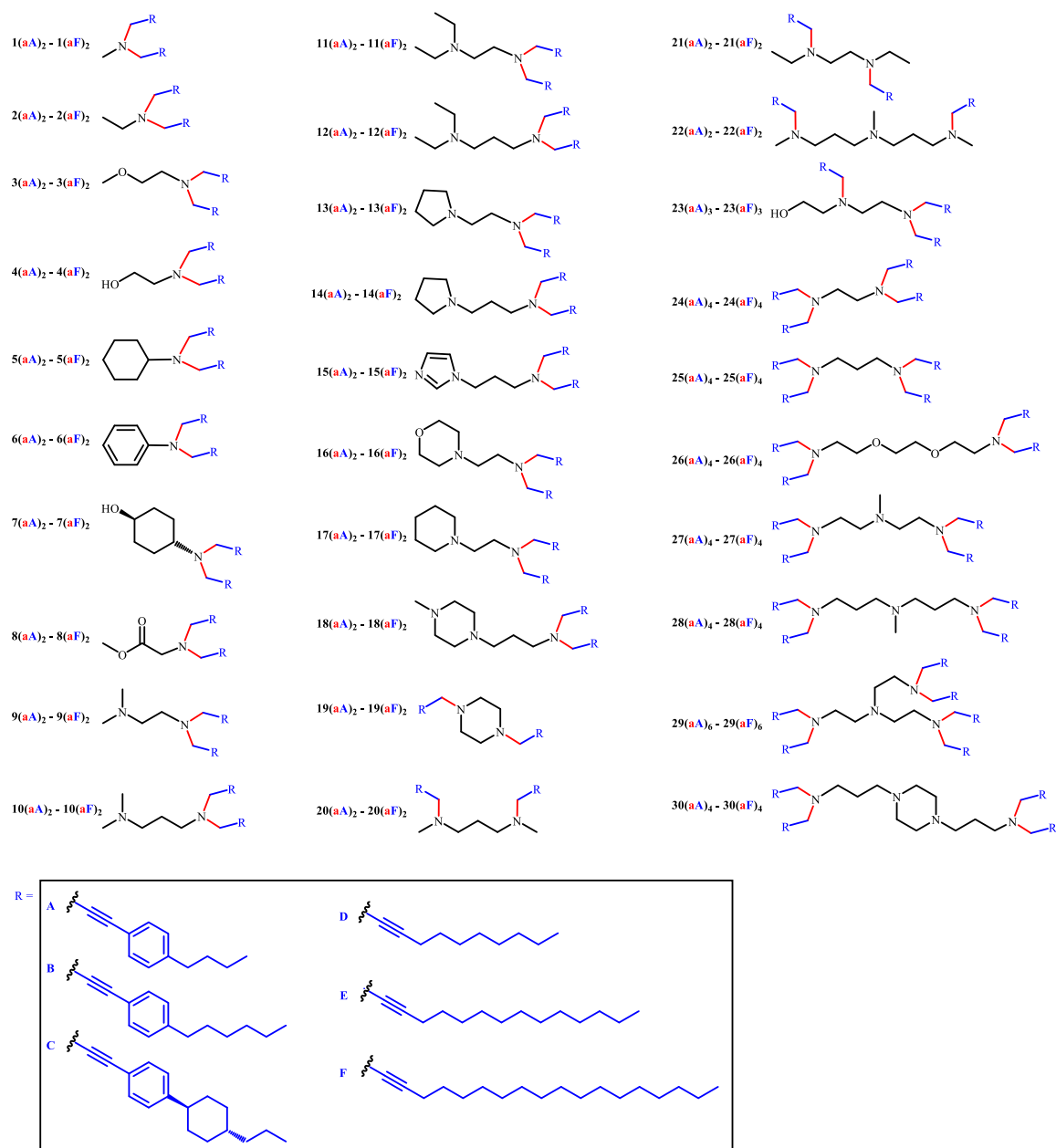
h: Brown oil, yield 91%. MS-ESI: calculated for $C_{19}H_{28}O_3$: 304.20, found $[M+H]^+ = 305.18$. 1H NMR (400 MHz, $CDCl_3$) δ 10.11 (s, 1H), 8.24 – 8.18 (m, 2H), 8.00 – 7.91 (m, 2H), 5.10 (ddd, $J = 12.6, 6.9, 5.6$ Hz, 1H), 1.79 – 1.58 (m, 4H), 1.45 – 1.17 (m, 12H), 0.96 (t, $J = 7.4$ Hz, 3H), 0.92 – 0.82 (m, 3H).

i: Colorless oil, yield 89%. MS-ESI: calculated for $C_{25}H_{40}O_3$: 388.30, found $[M+H]^+ = 389.31$. 1H NMR (400 MHz, $CDCl_3$) δ 10.11 (s, 1H), 8.23 – 8.16 (m, 2H), 7.99 – 7.91 (m, 2H), 5.15 (tt, $J = 7.1, 5.3$ Hz, 1H), 1.77 – 1.58 (m, 4H), 1.41 – 1.16 (m, 24H), 0.91 – 0.83 (m, 6H).



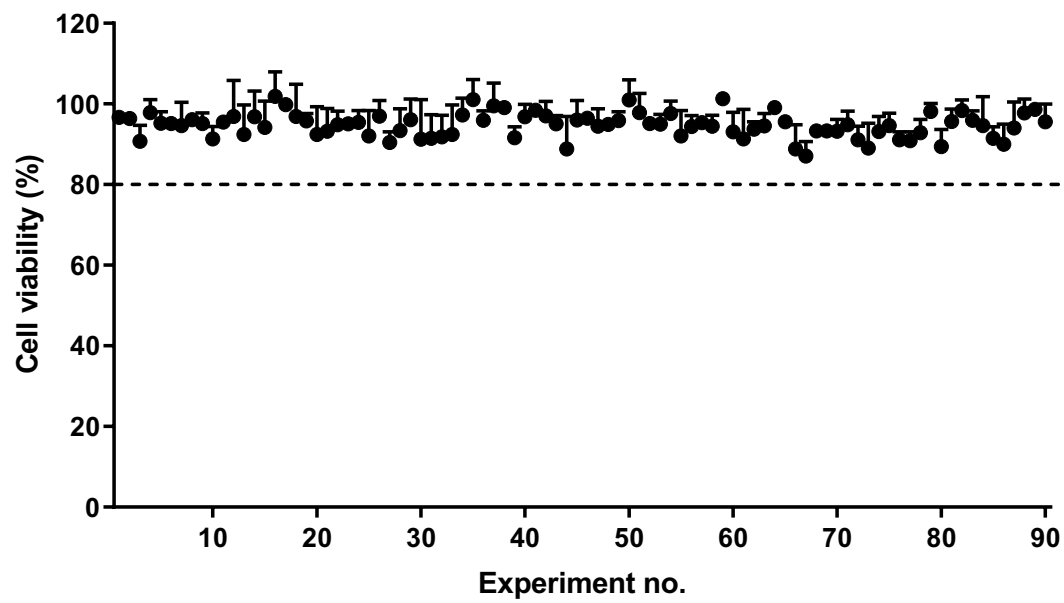
Supplementary Method 2 I Synthesis of the metabolite of 31hP (31hP-M). Amine 31 (0.25 mmol, 1.25 eq.), 4-formylbenzoic acid (0.2 mmol, 1 eq.), 5-hexynoic acid (0.2 mmol, 1 eq.) and copper(I) chloride (0.2 mmol, 0.1 eq.) were combined in a glass vial and stirred at 50 °C for 48 h. The crude mixture was passed through a silica plug using dichloromethane as the solvent, and the resulting filtrate was subsequently concentrated under reduced pressure. The crude product was loaded onto a silica column and subjected to column chromatography using a 10:90 MeOH/DCM eluent, resulting in the isolation of a brown liquid. The solvent was removed under vacuum to yield purified product.

31hP-M: Brown oil, yield 79%. MS-ESI: calculated for $C_{21}H_{30}N_2O_4$: 374.48, found $[M+H]^+ = 375.38$. 1H NMR (400 MHz, $CDCl_3$) δ 7.95-7.35 (m, 4H), 4.65 (s, 1H), 3.15-2.95 (m, 10H), 2.48-2.38 (m, 2H), 2.27 (s, 3H), 1.98-1.86 (m, 2H), 1.27-1.22 (m, 6H).

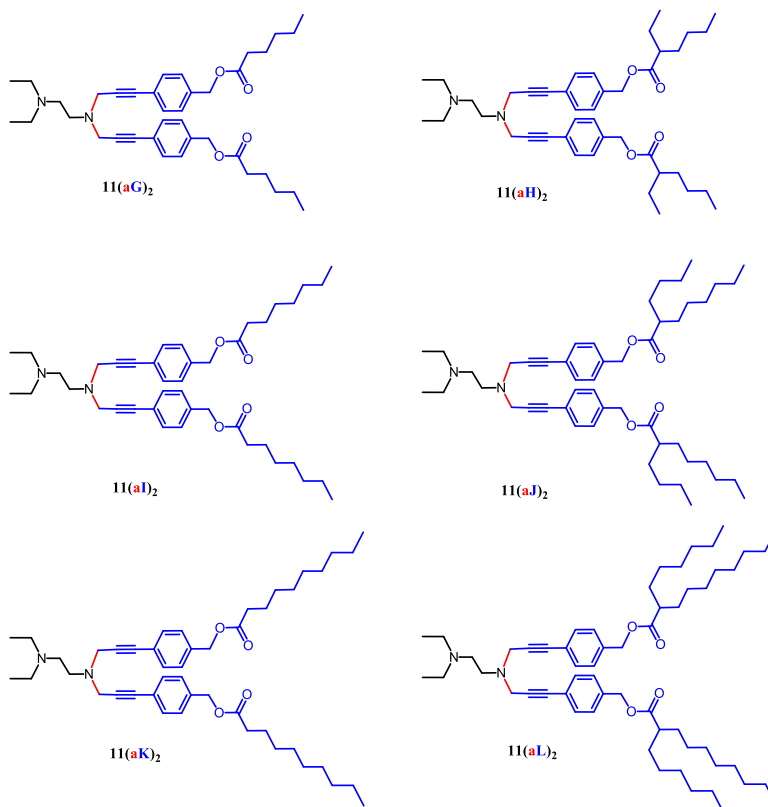


Supplementary Fig. 1 | Chemical structures of A³-lipids in Library 1. These A³-lipids were synthesized based on **Method A**. Crude A³-lipids were directly used for *in vitro* screening. The lead A³-lipid 11(aB)₂ was purified and characterized before *in vivo* use.

11(aB)₂: pale yellow oil, yield 53%. MS-ESI: calculated for C₃₆H₅₂N₂: 512.83, found [M+H]⁺ = 513.61. ¹H NMR (400 MHz, CDCl₃) δ 7.44 – 7.38 (m, 4H), 7.28 (d, J = 8.1 Hz, 4H), 3.75 (s, 4H), 2.35 (td, J = 7.5, 4.6 Hz, 4H), 2.15 – 2.03 (m, 8H), 1.71 – 1.46 (m, 4H), 1.30 (td, J = 5.9, 2.1 Hz, 12H), 1.13 – 0.98 (m, 6H), 0.90 (t, J = 6.6 Hz, 6H).



Supplementary Fig. 2 | Cell viability of A³-LNPs in Library 1. HepG2 cells were treated with various LNPs at a dose of 15 ng mRNA/well for 24 h. No major cytotoxicity was observed for each LNP. The dashed line indicates 80% cell viability. Data are presented as mean $\pm$ SD (n=2 independent biological replicates, initial screening).



Supplementary Fig. 3 I Chemical structures of A³-lipids in Library 2. These A³-lipids were synthesized based on **Method A**, which were purified and characterized before use.

11(aG)₂: pale yellow oil, yield 55%. MS-ESI: calculated for C₃₈H₅₂N₂O₄: 600.84, found [M+H]⁺ = 601.72. ¹H NMR (400 MHz, CDCl₃) δ 7.45 – 7.38 (m, 4H), 7.27 (d, J = 7.9 Hz, 4H), 5.09 (s, 4H), 3.75 (s, 4H), 2.85 – 2.54 (m, 8H), 2.35 (t, J = 7.5 Hz, 4H), 1.65 (q, J = 7.4 Hz, 4H), 1.31 (td, J = 5.9, 2.1 Hz, 8H), 1.06 (t, J = 7.2 Hz, 6H), 0.93 – 0.84 (m, 6H).

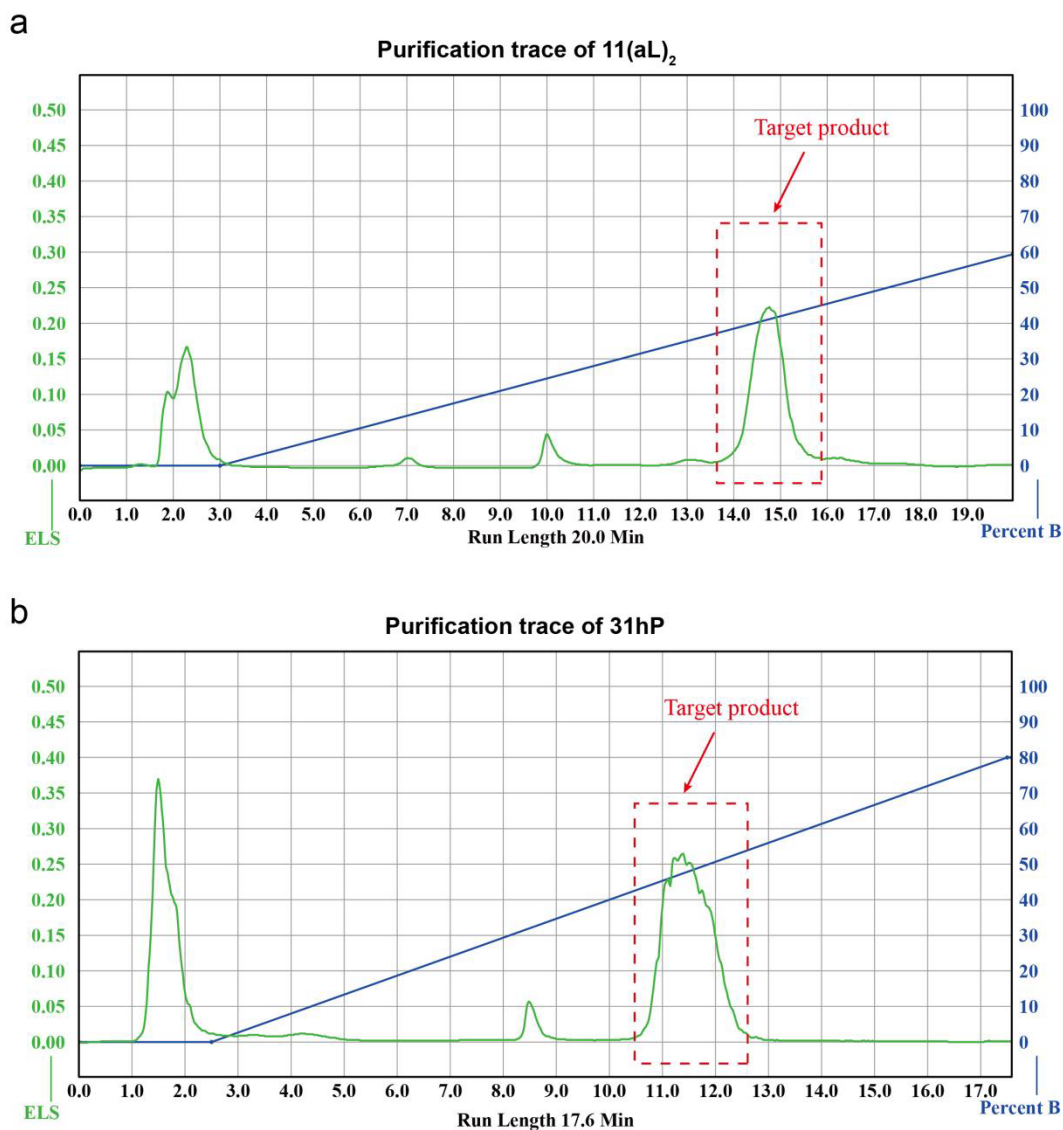
11(aH)₂: pale yellow oil, yield 47%. MS-ESI: calculated for C₄₂H₆₀N₂O₄: 656.95, found [M+H]⁺ = 657.70. ¹H NMR (400 MHz, CDCl₃) δ 7.43 – 7.39 (m, 4H), 7.28 (d, J = 8.2 Hz, 4H), 5.10 (s, 4H), 3.75 (s, 4H), 2.86 – 2.49 (m, 8H), 2.37 – 2.26 (m, 2H), 1.71 – 1.40 (m, 8H), 1.36 – 1.15 (m, 8H), 1.13 – 0.98 (m, 6H), 0.91 – 0.83 (m, 12H).

11(aI)₂: pale yellow oil, yield 54%. MS-ESI: calculated for C₄₂H₆₀N₂O₄: 656.95, found [M+H]⁺ = 657.72. ¹H NMR (400 MHz, CDCl₃) δ 7.45 – 7.38 (m, 4H), 7.28 (d, J = 1.8 Hz, 4H), 5.09 (s, 4H), 3.75 (s, 4H), 2.88 – 2.52 (m, 8H), 2.35 (t, J = 7.5 Hz, 4H), 1.63 (d, J = 7.5 Hz, 4H), 1.39 – 1.22 (m, 16H), 1.13 – 0.97 (m, 6H), 0.91 – 0.83 (m, 6H).

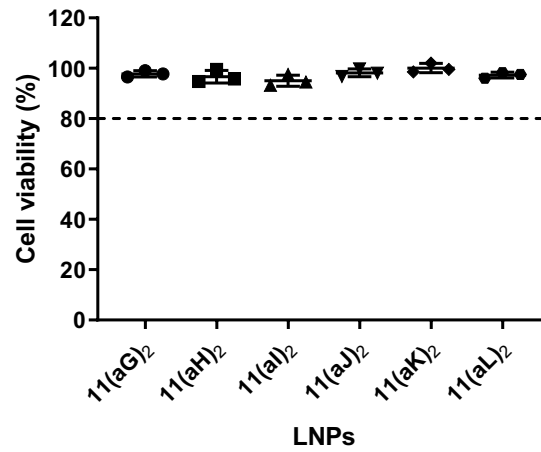
11(aJ)₂: pale yellow oil, yield 50%. MS-ESI: calculated for C₅₀H₇₆N₂O₄: 769.17, found [M+H]⁺ = 770.12. ¹H NMR (400 MHz, CDCl₃) δ 7.45 – 7.37 (m, 4H), 7.31 – 7.25 (m, 4H), 5.10 (s, 4H), 3.75 (s, 4H), 2.90 – 2.52 (m, 8H), 2.35 – 2.23 (m, 2H), 1.68 – 1.38 (m, 8H), 1.29 – 1.17 (m, 24H), 1.12 – 1.00 (m, 6H), 0.91 – 0.82 (m, 12H).

11(aK)₂: pale yellow oil, yield 51%. MS-ESI: calculated for C₄₆H₆₈N₂O₄: 713.06, found [M+H]⁺ = 713.94. ¹H NMR (400 MHz, CDCl₃) δ 7.45 – 7.38 (m, 4H), 7.27 (d, J = 7.9 Hz, 4H), 5.09 (s, 4H), 3.75 (s, 4H), 2.87 – 2.52 (m, 8H), 2.35 (t, J = 7.5 Hz, 4H), 1.64 (d, J = 7.5 Hz, 4H), 1.39 – 1.20 (m, 24H), 1.12 – 0.98 (m, 6H), 0.93 – 0.83 (m, 6H).

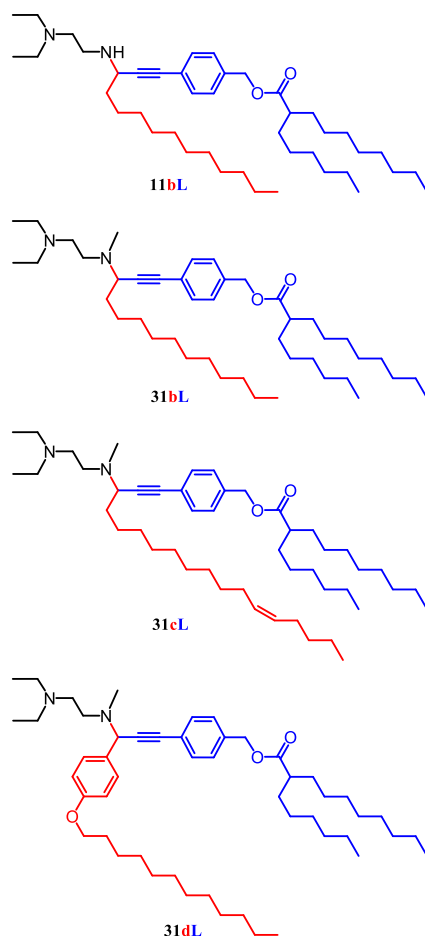
11(aL)₂: bright yellow oil, yield 48%. MS-ESI: calculated for C₅₈H₉₂N₂O₄: 881.38, found [M+H]⁺ = 882.09. ¹H NMR (400 MHz, CDCl₃) δ 7.45 – 7.37 (m, 4H), 7.32 – 7.26 (m, 4H), 5.10 (s, 4H), 3.75 (s, 4H), 2.91 – 2.56 (m, 8H), 2.37 (tt, J = 8.9, 5.3 Hz, 2H), 1.60 (dd, J = 13.9, 8.8 Hz, 4H), 1.51 – 1.38 (dd, J = 13.9, 8.8 Hz, 4H), 1.27 – 1.19 (m, 40H), 1.12 – 1.01 (m, 6H), 0.91 – 0.82 (m, 12H).



Supplementary Fig. 4 | Representative purification traces of A³-lipids. **a**, The purification trace of 11(aL)₂ recorded by a CombiFlash NextGen 300+ chromatography system equipped with evaporative light scattering (ELS) detectors. The crude product was purified with gradient elution from 100% DCM to 60% DCM/MeOH/NH₄OH (75:22:3, aq). **b**, The purification trace of 31hP recorded by a CombiFlash NextGen 300+ chromatography system equipped with ELS detectors. The crude product was purified with gradient elution from 100% DCM to 80% DCM/MeOH/NH₄OH (75:22:3, aq).



Supplementary Fig. 5 I Cell viability of A³-LNPs in Library 2. HepG2 cells were treated with various LNPs at a dose of 15 ng mRNA/well for 24 h. No major cytotoxicity was observed for each LNP. The dashed line indicates 80% cell viability. Data are presented as mean ± SD (n=3 independent biological replicates).



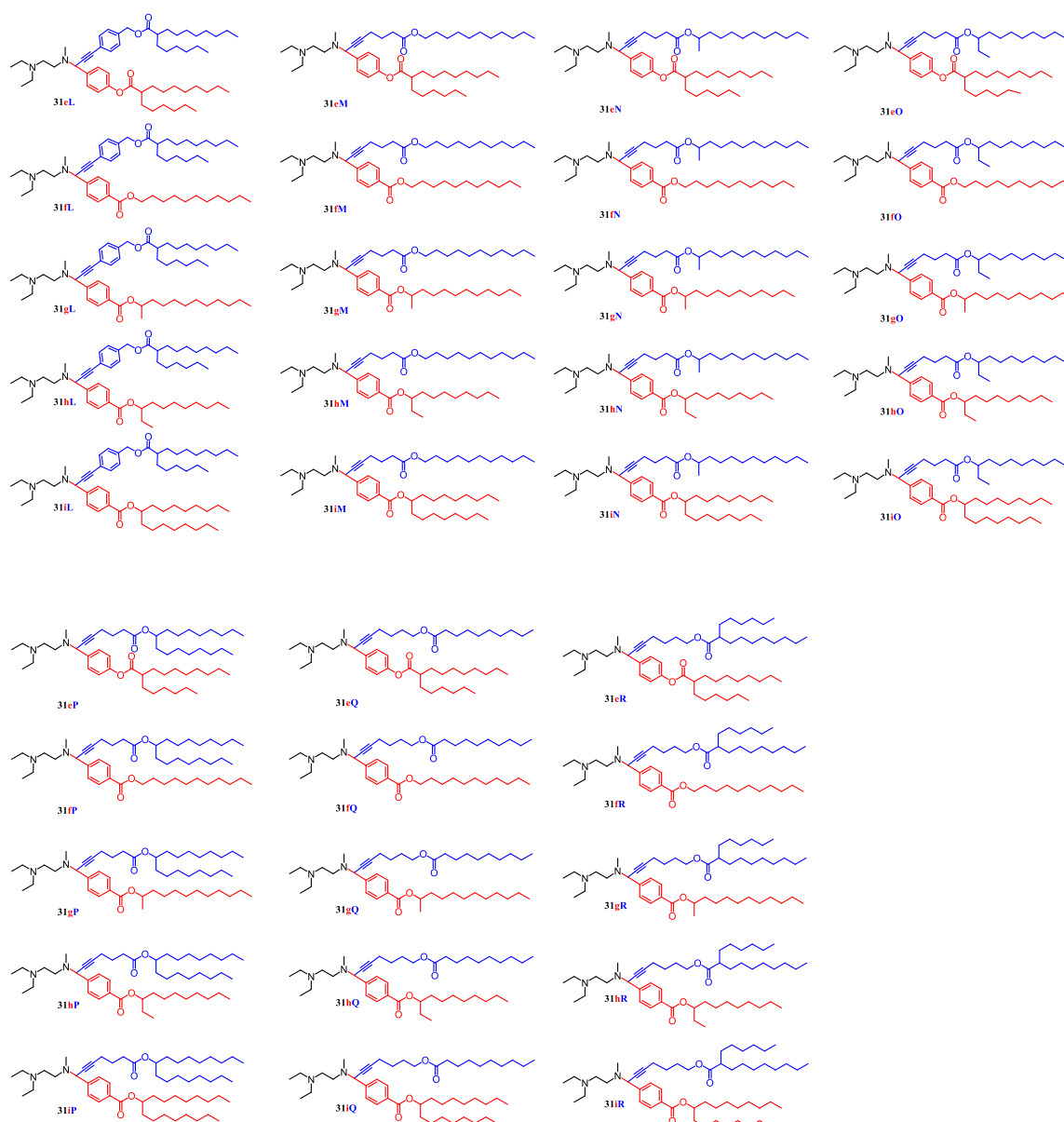
Supplementary Fig. 6 I Chemical structures of A³-lipids in Library 3. These A³-lipids were synthesized based on **Method B**, which were purified and characterized before use.

11bL: pale yellow oil, yield 62%. MS-ESI: calculated for C₄₃H₇₆N₂O₂: 653.09, found [M+H]⁺ = 653.91. ¹H NMR (400 MHz, CDCl₃) δ 7.43 (d, J = 8.2 Hz, 2H), 7.30 (d, J = 8.8 Hz, 2H), 5.12 (s, 2H), 3.62 (t, J = 7.4 Hz, 1H), 2.82 – 2.51 (m, 8H), 2.40 (tt, J = 8.7, 5.4 Hz, 1H), 1.74 – 1.40 (m, 6H), 1.29 (t, J = 14.2 Hz, 38H), 1.10 – 0.85 (m, 15H).

31bL: pale yellow oil, yield 54%. MS-ESI: calculated for C₄₄H₇₈N₂O₂: 667.12, found [M+H]⁺ = 667.96. ¹H NMR (400 MHz, CDCl₃) δ 7.42 (d, J = 8.2 Hz, 2H), 7.30 (d, J = 8.8 Hz, 2H), 5.12 (s, 2H), 3.62 (t, J = 7.5 Hz, 1H), 2.80 – 2.50 (m, 8H), 2.39 (tt, J = 8.8, 5.4 Hz, 1H), 2.34 (s, 3H), 1.75 – 1.55 (m, 4H), 1.57 – 1.41 (m, 2H), 1.29 (t, J = 14.3 Hz, 38H), 1.09 (s, 6H), 0.94 – 0.85 (m, 9H).

31cL: pale yellow oil, yield 51%. MS-ESI: calculated for C₄₈H₈₄N₂O₂: 721.21, found [M+H]⁺ = 721.97. ¹H NMR (400 MHz, CDCl₃) δ 7.45 – 7.39 (m, 2H), 7.30 (d, J = 8.8 Hz, 2H), 5.37 (td, J = 4.4, 2.2 Hz, 2H), 5.12 (s, 2H), 3.62 (t, J = 7.4 Hz, 1H), 2.62 (s, 8H), 2.44 – 2.36 (m, 1H), 2.34 (s, 3H), 2.04 (t, J = 4.7 Hz, 4H), 1.75 – 1.41 (m, 6H), 1.38 – 1.19 (m, 38H), 1.08 (s, 6H), 0.91 (qd, J = 7.1, 3.3 Hz, 9H).

31dL: brown oil, yield 43%. MS-ESI: calculated for C₅₁H₈₄N₂O₃: 773.24, found [M+H]⁺ = 774.18. ¹H NMR (400 MHz, CDCl₃) δ 7.57 – 7.28 (m, 6H), 6.96 – 6.87 (m, 2H), 5.14 (s, 2H), 4.95 (s, 1H), 3.99 (dt, J = 13.2, 6.4 Hz, 2H), 3.05 – 2.72 (m, 8H), 2.61 – 2.49 (m, 1H), 2.34 (s, 3H), 1.80 (p, J = 6.6 Hz, 2H), 1.63 – 1.31 (m, 4H), 1.28 (d, J = 11.4 Hz, 38H), 1.10 – 0.84 (m, 15H).



Supplementary Fig. 7 I Chemical structures of A³-lipids in Library 4. These A³-lipids were synthesized based on **Method B**, which were purified and characterized before use.

31eL: brown oil, yield 56%. MS-ESI: calculated for C₅₅H₉₀N₂O₄: 843.34, found [M+H]⁺ = 844.21. ¹H NMR (400 MHz, CDCl₃) δ 7.67 – 7.60 (m, 2H), 7.52 – 7.45 (m, 2H), 7.35 – 7.29 (m, 2H), 7.08 – 7.00 (m, 2H), 5.12 (s, 2H), 5.00 (s, 1H), 2.56 (t, J = 7.2 Hz, 8H), 2.27 (s, 3H), 1.75 (dq, J = 15.4, 7.7 Hz, 2H), 1.49 – 1.17 (m, 48H), 1.08 – 0.98 (m, 6H), 0.93 – 0.83 (m, 12H).

31fL: pale yellow oil, yield 54%. MS-ESI: calculated for C₅₁H₈₂N₂O₄: 787.23, found [M+H]⁺ = 788.02. ¹H NMR (400 MHz, CDCl₃) δ 7.67 – 7.60 (m, 2H), 7.53 – 7.46 (m, 2H), 7.35 – 7.28 (m, 2H), 7.08 – 7.01 (m, 2H), 5.12 (s, 2H), 4.98 (s, 1H), 4.11 (t, 2H), 2.54 (m, 8H), 2.17 (s, 3H), 1.97 (m, 1H), 1.76 – 1.81 (m, 2H), 1.59 – 1.48 (m, 4H), 1.45 – 1.16 (m, 36H), 1.08 – 0.97 (m, 6H), 0.92 – 0.81 (m, 9H).

31gL: yellow oil, yield 47%. MS-ESI: calculated for C₅₂H₈₄N₂O₄: 801.25, found [M+H]⁺ = 802.03. ¹H NMR (400 MHz, CDCl₃) δ 7.67 – 7.61 (m, 2H), 7.52 – 7.45 (m, 2H), 7.35 – 7.28 (m, 2H), 7.08 – 7.00 (m, 2H), 5.11 (s, 2H), 5.15 – 5.06 (m, 1H), 4.97 (s, 1H), 2.65 – 2.41 (m, 8H), 2.16 (s, 3H), 1.97 (m, 1H), 1.68 – 1.49 (m, 6H), 1.42 – 1.18 (m, 39H), 1.08 – 0.97 (m, 6H), 0.92 – 0.82 (m, 9H).

31hL: yellow oil, yield 52%. MS-ESI: calculated for C₅₁H₈₂N₂O₄: 787.23, found [M+H]⁺ = 788.03. ¹H NMR (400 MHz, CDCl₃) δ 7.67 – 7.60 (m, 2H), 7.51 – 7.45 (m, 2H), 7.35 – 7.28 (m, 2H), 7.08 – 7.00 (m, 2H), 5.11

(s, 2H), 5.15 – 5.05 (m, 1H), 4.98 (s, 1H), 2.66 – 2.42 (m, 8H), 2.17 (s, 3H), 1.98 (m, 1H), 1.67 – 1.47 (m, 8H), 1.42 – 1.18 (m, 32H), 1.08 – 0.96 (m, 6H), 0.93 – 0.83 (m, 12H).

31iL: brown oil, yield 42%. MS-ESI: calculated for $C_{57}H_{94}N_2O_4$: 871.39, found $[M+H]^+ = 872.25$. 1H NMR (400 MHz, $CDCl_3$) δ 7.67 – 7.61 (m, 2H), 7.52 – 7.44 (m, 2H), 7.36 – 7.29 (m, 2H), 7.09 – 7.01 (m, 2H), 5.12 (s, 2H), 5.15 – 5.05 (m, 1H), 4.97 (s, 1H), 2.68 – 2.43 (m, 8H), 2.19 (s, 3H), 1.96 (m, 1H), 1.67 – 1.47 (m, 8H), 1.44 – 1.17 (m, 44H), 1.08 – 0.97 (m, 6H), 0.92 – 0.81 (m, 12H).

31eM: brown oil, yield 54%. MS-ESI: calculated for $C_{47}H_{82}N_2O_4$: 739.18, found $[M+H]^+ = 740.02$. 1H NMR (400 MHz, $CDCl_3$) δ 7.58 (d, $J = 8.5$ Hz, 2H), 7.05 (d, $J = 8.5$ Hz, 2H), 4.86 (s, 1H), 4.63 (t, $J = 7.0$ Hz, 2H), 2.71 – 2.51 (m, 8H), 2.40 (t, $J = 7.0$ Hz, 2H), 2.26 (s, 3H), 2.12 (m, 1H), 1.91 (t, $J = 7.1$ Hz, 2H), 1.76 (m, 2H), 1.62 – 1.48 (m, 6H), 1.42 – 1.16 (m, 36H), 1.08 – 0.97 (m, 6H), 0.94 – 0.81 (m, 9H).

31fM: brown oil, yield 56%. MS-ESI: calculated for $C_{43}H_{74}N_2O_4$: 683.08, found $[M+H]^+ = 683.91$. 1H NMR (400 MHz, $CDCl_3$) δ 7.99 (d, $J = 8.5$ Hz, 2H), 7.64 (d, $J = 8.5$ Hz, 2H), 4.88 (s, 1H), 4.68 – 4.22 (m, 4H), 2.71 – 2.52 (m, 8H), 2.41 (t, $J = 7.1$ Hz, 2H), 2.26 (s, 3H), 1.92 – 1.55 (m, 8H), 1.43 – 1.16 (m, 32H), 1.08 – 0.97 (m, 6H), 0.92 – 0.81 (m, 6H).

31gM: brown oil, yield 48%. MS-ESI: calculated for $C_{44}H_{76}N_2O_4$: 697.10, found $[M+H]^+ = 697.97$. 1H NMR (400 MHz, $CDCl_3$) δ 7.97 (d, $J = 8.4$ Hz, 2H), 7.63 (d, $J = 8.5$ Hz, 2H), 5.04 (p, $J = 6.1$ Hz, 1H), 4.84 (s, 1H), 4.12 (t, $J = 6.5$ Hz, 2H), 2.72 – 2.52 (m, 8H), 2.42 (t, $J = 7.1$ Hz, 2H), 2.24 (s, 3H), 1.95 – 1.53 (m, 8H), 1.48 – 1.21 (m, 35H), 1.06 – 0.95 (m, 6H), 0.93 – 0.82 (m, 6H).

31hM: yellow oil, yield 49%. MS-ESI: calculated for $C_{43}H_{74}N_2O_4$: 683.08, found $[M+H]^+ = 683.95$. 1H NMR (400 MHz, $CDCl_3$) δ 7.98 (d, $J = 8.5$ Hz, 2H), 7.63 (d, $J = 8.5$ Hz, 2H), 5.06 (m, 1H), 4.83 (s, 1H), 4.11 (t, $J = 6.6$ Hz, 2H), 2.72 – 2.51 (m, 8H), 2.43 (t, $J = 6.8$ Hz, 2H), 2.22 (s, 3H), 1.97 – 1.52 (m, 10H), 1.45 – 1.22 (m, 28H), 1.07 – 0.95 (m, 6H), 0.92 – 0.82 (m, 9H).

31iM: brown oil, yield 53%. MS-ESI: calculated for $C_{49}H_{86}N_2O_4$: 767.24, found $[M+H]^+ = 768.08$. 1H NMR (400 MHz, $CDCl_3$) δ 7.99 (d, $J = 8.5$ Hz, 2H), 7.63 (d, $J = 8.4$ Hz, 2H), 5.05 (m, 1H), 4.83 (s, 1H), 4.12 (t, $J = 6.5$ Hz, 2H), 2.73 – 2.52 (m, 8H), 2.42 (t, $J = 6.7$ Hz, 2H), 2.20 (s, 3H), 1.96 – 1.52 (m, 10H), 1.47 – 1.22 (m, 40H), 1.08 – 0.97 (m, 6H), 0.92 – 0.81 (m, 9H).

31eN: pale yellow oil, yield 50%. MS-ESI: calculated for $C_{48}H_{84}N_2O_4$: 753.21, found $[M+H]^+ = 754.12$. 1H NMR (400 MHz, $CDCl_3$) δ 7.57 (d, $J = 8.5$ Hz, 2H), 7.04 (d, $J = 8.4$ Hz, 2H), 5.14 – 5.07 (m, 1H), 4.82 (s, 1H), 2.68 – 2.44 (m, 8H), 2.40 – 2.15 (m, 6H), 1.95 – 1.78 (m, 4H), 1.61 – 1.47 (m, 6H), 1.42 – 1.19 (m, 39H), 0.99 (t, $J = 7.2$ Hz, 6H), 0.91 – 0.80 (m, 9H).

31fN: pale yellow oil, yield 52%. MS-ESI: calculated for $C_{44}H_{76}N_2O_4$: 697.10, found $[M+H]^+ = 697.95$. 1H NMR (400 MHz, $CDCl_3$) δ 7.88 (d, $J = 8.5$ Hz, 2H), 7.02 (d, $J = 8.5$ Hz, 2H), 5.02 (m, 1H), 4.88 (s, 1H), 4.65 – 4.23 (t, $J = 7.1$ Hz, 2H), 2.72 – 2.52 (m, 8H), 2.42 (t, $J = 7.1$ Hz, 2H), 2.23 (s, 3H), 1.93 – 1.59 (m, 8H), 1.49 – 1.18 (m, 35H), 1.08 – 0.96 (m, 6H), 0.92 – 0.82 (m, 6H).

31gN: pale yellow oil, yield 43%. MS-ESI: calculated for $C_{45}H_{78}N_2O_4$: 711.13, found $[M+H]^+ = 711.98$. 1H NMR (400 MHz, $CDCl_3$) δ 8.01 (d, $J = 8.4$ Hz, 2H), 7.62 (d, $J = 8.4$ Hz, 2H), 5.03 (m, 2H), 4.86 (s, 1H), 2.73 – 2.52 (m, 8H), 2.43 (t, $J = 7.2$ Hz, 2H), 2.23 (s, 3H), 1.94 – 1.75 (m, 4H), 1.68 – 1.45 (m, 4H), 1.33 – 1.19 (m, 38H), 1.07 – 0.96 (m, 6H), 0.92 – 0.82 (m, 6H).

31hN: pale yellow oil, yield 41%. MS-ESI: calculated for $C_{44}H_{76}N_2O_4$: 697.10, found $[M+H]^+ = 697.96$. 1H NMR (400 MHz, $CDCl_3$) δ 7.98 (d, $J = 8.5$ Hz, 2H), 7.62 (d, $J = 8.4$ Hz, 2H), 5.06 (p, $J = 6.2$ Hz, 1H), 4.89 (p, $J = 6.2$ Hz, 1H), 4.81 (s, 1H), 2.71 – 2.52 (m, 8H), 2.42 (t, $J = 7.1$ Hz, 2H), 2.20 (s, 3H), 1.94 – 1.48 (m, 10H), 1.35 – 1.18 (m, 31H), 1.08 – 0.96 (m, 6H), 0.91 – 0.80 (m, 9H).

31iN: pale yellow oil, yield 52%. MS-ESI: calculated for $C_{50}H_{88}N_2O_4$: 781.26, found $[M+H]^+ = 782.11$. 1H NMR (400 MHz, $CDCl_3$) δ 7.99 (d, 2H), 7.64 (d, $J = 8.2$ Hz, 2H), 5.16 – 5.06 (m, 1H), 4.98 – 4.86 (m, 1H), 4.78 (s, 1H), 2.67 – 2.43 (m, 8H), 2.41 (td, $J = 7.0, 2.1$ Hz, 2H), 2.17 (s, 3H), 1.90 (p, $J = 7.1$ Hz, 2H), 1.65 (p, $J = 7.7$ Hz, 2H), 1.40 – 1.17 (m, 49H), 0.99 (t, $J = 7.1$ Hz, 6H), 0.91 – 0.82 (m, 9H).

31eO: brown oil, yield 52%. MS-ESI: calculated for $C_{47}H_{82}N_2O_4$: 739.18, found $[M+H]^+ = 740.01$. 1H NMR (400 MHz, $CDCl_3$) δ 7.59 (d, $J = 8.5$ Hz, 2H), 7.04 (d, $J = 8.5$ Hz, 2H), 4.86 (t, $J = 6.2$ Hz, 1H), 4.76 (s, 1H),

2.74 – 2.53 (m, 8H), 2.54 – 2.38 (m, 3H), 2.22 (s, 3H), 1.93 (p, $J = 7.1$ Hz, 2H), 1.67 – 1.54 (m, 10H), 1.27 (s, 32H), 1.07 (t, $J = 7.1$ Hz, 6H), 0.95 – 0.85 (m, 12H).

31fO: pale yellow oil, yield 46%. MS-ESI: calculated for $C_{43}H_{74}N_2O_4$: 683.08, found $[M+H]^+ = 683.92$. 1H NMR (400 MHz, $CDCl_3$) δ 8.02 (d, $J = 8.5$ Hz, 2H), 7.63 (d, $J = 8.5$ Hz, 2H), 5.03 (m, 1H), 4.85 (s, 1H), 4.36 (t, $J = 7.2$ Hz, 2H), 2.71 – 2.52 (m, 8H), 2.42 (t, $J = 7.0$ Hz, 2H), 2.25 (s, 3H), 1.97 – 1.52 (m, 10H), 1.43 – 1.18 (m, 28H), 1.05 (t, $J = 7.2$ Hz, 6H), 0.95 – 0.82 (m, 9H).

31gO: yellow oil, yield 47%. MS-ESI: calculated for $C_{44}H_{76}N_2O_4$: 697.10, found $[M+H]^+ = 697.95$. 1H NMR (400 MHz, $CDCl_3$) δ 7.98 (d, $J = 8.5$ Hz, 2H), 7.65 (d, $J = 8.5$ Hz, 2H), 5.03 (m, 2H), 4.88 (s, 1H), 2.73 – 2.51 (m, 8H), 2.42 (t, $J = 7.2$ Hz, 2H), 2.24 (s, 3H), 1.95 – 1.75 (m, 4H), 1.69 – 1.43 (m, 6H), 1.34 – 1.17 (m, 31H), 1.08 – 0.96 (m, 6H), 0.92 – 0.82 (m, 9H).

31hO: yellow oil, yield 46%. MS-ESI: calculated for $C_{43}H_{74}N_2O_4$: 683.08, found $[M+H]^+ = 683.96$. 1H NMR (400 MHz, $CDCl_3$) δ 7.99 (d, $J = 8.4$ Hz, 2H), 7.63 (d, $J = 8.5$ Hz, 2H), 5.02 (m, 2H), 4.87 (s, 1H), 2.71 – 2.52 (m, 8H), 2.40 (t, $J = 7.1$ Hz, 2H), 2.23 (s, 3H), 1.96 – 1.75 (m, 4H), 1.68 – 1.41 (m, 8H), 1.34 – 1.15 (m, 24H), 1.02 (t, $J = 7.2$ Hz, 6H), 0.92 – 0.81 (m, 12H).

31iO: pale yellow oil, yield 43%. MS-ESI: calculated for $C_{49}H_{86}N_2O_4$: 767.24, found $[M+H]^+ = 768.14$. 1H NMR (400 MHz, $CDCl_3$) δ 8.03 – 7.96 (m, 2H), 7.68 – 7.60 (m, 2H), 5.11 (p, $J = 6.2$ Hz, 1H), 4.89 – 4.76 (m, 2H), 2.51 (dt, $J = 14.8, 7.1$ Hz, 8H), 2.41 (td, $J = 7.0, 2.1$ Hz, 2H), 2.17 (s, 3H), 1.91 (p, $J = 7.2$ Hz, 2H), 1.71 – 1.48 (m, 10H), 1.26 (d, $J = 9.4$ Hz, 36H), 1.00 (t, $J = 7.1$ Hz, 6H), 0.94 – 0.75 (m, 12H).

31eP: brown oil, yield 44%. MS-ESI: calculated for $C_{53}H_{94}N_2O_4$: 823.35, found $[M+H]^+ = 824.13$. 1H NMR (400 MHz, $CDCl_3$) δ 7.59 (d, $J = 8.4$ Hz, 2H), 7.04 (d, $J = 8.5$ Hz, 2H), 4.91 (t, $J = 6.3$ Hz, 1H), 4.76 (s, 1H), 2.59 (p, $J = 7.1$ Hz, 8H), 2.53 – 2.38 (m, 3H), 2.22 (s, 3H), 1.98 – 1.87 (m, 2H), 1.69 – 1.49 (m, 10H), 1.46 – 1.18 (m, 44H), 1.07 (t, $J = 7.1$ Hz, 6H), 0.90 (tt, $J = 7.3, 2.6$ Hz, 12H).

31fP: brown oil, yield 48%. MS-ESI: calculated for $C_{49}H_{86}N_2O_4$: 767.24, found $[M+H]^+ = 768.05$. 1H NMR (400 MHz, $CDCl_3$) δ 7.98 (d, $J = 8.5$ Hz, 2H), 7.62 (d, $J = 8.5$ Hz, 2H), 5.02 (m, 1H), 4.86 (s, 1H), 4.37 (t, $J = 7.1$ Hz, 2H), 2.72 – 2.51 (m, 8H), 2.43 (t, $J = 7.1$ Hz, 2H), 2.26 (s, 3H), 1.97 – 1.52 (m, 10H), 1.44 – 1.17 (m, 40H), 1.04 (t, $J = 7.2$ Hz, 6H), 0.95 – 0.83 (m, 9H).

31gP: brown oil, yield 45%. MS-ESI: calculated for $C_{50}H_{88}N_2O_4$: 781.26, found $[M+H]^+ = 782.12$. 1H NMR (400 MHz, $CDCl_3$) δ 8.00 (d, $J = 8.5$ Hz, 2H), 7.65 (d, $J = 8.4$ Hz, 2H), 5.06 (m, 1H), 4.91 (m, 1H), 4.84 (s, 1H), 2.70 – 2.52 (m, 8H), 2.41 (t, $J = 7.0$ Hz, 2H), 2.23 (s, 3H), 1.98 – 1.78 (m, 4H), 1.64 – 1.48 (m, 6H), 1.27 – 1.15 (m, 43H), 1.04 (t, $J = 7.1$ Hz, 6H), 0.96 – 0.81 (m, 9H).

31hP: brown oil, yield 51%. MS-ESI: calculated for $C_{49}H_{86}N_2O_4$: 767.24, found $[M+H]^+ = 768.03$. 1H NMR (400 MHz, $CDCl_3$) δ 8.04 – 7.96 (m, 2H), 7.64 (d, $J = 8.2$ Hz, 2H), 5.06 (p, $J = 6.1$ Hz, 1H), 4.89 (p, $J = 6.2$ Hz, 1H), 4.79 (s, 1H), 2.69 – 2.43 (m, 8H), 2.41 (td, $J = 7.0, 2.1$ Hz, 2H), 2.17 (s, 3H), 1.91 (p, $J = 7.2$ Hz, 2H), 1.75 – 1.60 (m, 4H), 1.52 (d, $J = 6.4$ Hz, 4H), 1.26 (d, $J = 6.9$ Hz, 38H), 1.00 (t, $J = 7.1$ Hz, 6H), 0.94 (td, $J = 7.4, 0.9$ Hz, 3H), 0.87 (td, $J = 6.9, 2.4$ Hz, 9H).

31iP: brown oil, yield 48%. MS-ESI: calculated for $C_{55}H_{98}N_2O_4$: 851.40, found $[M+H]^+ = 852.38$. 1H NMR (400 MHz, $CDCl_3$) δ 7.99 (d, $J = 8.4$ Hz, 2H), 7.64 (d, $J = 8.4$ Hz, 2H), 5.08 (m, 2H), 4.87 (s, 1H), 2.72 – 2.52 (m, 8H), 2.43 (t, $J = 7.1$ Hz, 2H), 2.21 (s, 3H), 1.97 – 1.79 (m, 4H), 1.64 – 1.47 (m, 8H), 1.27 – 1.16 (m, 48H), 1.05 (t, $J = 7.3$ Hz, 6H), 0.95 – 0.83 (m, 12H).

31eQ: yellow oil, yield 53%. MS-ESI: calculated for $C_{46}H_{80}N_2O_4$: 725.16, found $[M+H]^+ = 726.01$. 1H NMR (400 MHz, $CDCl_3$) δ 7.83 – 7.76 (m, 2H), 7.01 – 6.95 (m, 2H), 4.75 (s, 1H), 4.13 (td, $J = 6.5, 3.0$ Hz, 2H), 2.69 – 2.50 (m, 8H), 2.43 – 2.21 (m, 6H), 1.85 – 1.51 (m, 10H), 4.45 – 1.20 (m, 34H), 1.05 (dt, $J = 16.8, 7.1$ Hz, 6H), 0.90 (td, $J = 6.0, 3.4$ Hz, 9H).

31fQ: light yellow oil, yield 58%. MS-ESI: calculated for $C_{42}H_{72}N_2O_4$: 669.05, found $[M+H]^+ = 669.97$. 1H NMR (400 MHz, $CDCl_3$) δ 7.98 (d, $J = 8.4$ Hz, 2H), 7.65 (d, $J = 8.5$ Hz, 2H), 4.83 (s, 1H), 4.38 (t, $J = 7.2$ Hz, 2H), 4.15 (t, $J = 6.5$ Hz, 2H), 2.71 – 2.52 (m, 8H), 2.42 (t, $J = 7.1$ Hz, 2H), 2.26 (s, 3H), 1.99 – 1.54 (m, 8H), 1.47 – 1.15 (m, 30H), 1.04 (t, $J = 7.4$ Hz, 6H), 0.89 (t, $J = 6.1$ Hz, 6H).

31gQ: light yellow oil, yield 54%. MS-ESI: calculated for $C_{43}H_{74}N_2O_4$: 683.08, found $[M+H]^+ = 683.98$. 1H NMR (400 MHz, $CDCl_3$) δ 7.99 (d, $J = 8.5$ Hz, 2H), 7.63 (d, $J = 8.5$ Hz, 2H), 5.04 (m, 1H), 4.82 (s, 1H), 4.16 (t, $J = 7.0$ Hz, 2H), 2.73 – 2.53 (m, 8H), 2.44 (t, $J = 7.2$ Hz, 2H), 2.24 (s, 3H), 1.97 – 1.56 (m, 8H), 1.48 – 1.16 (m, 33H), 1.02 (t, $J = 7.2$ Hz, 6H), 0.89 (t, $J = 6.2$ Hz, 6H).

31hQ: light yellow oil, yield 54%. MS-ESI: calculated for $C_{42}H_{72}N_2O_4$: 669.05, found $[M+H]^+ = 669.97$. 1H NMR (400 MHz, $CDCl_3$) δ 8.01 (d, $J = 8.5$ Hz, 2H), 7.65 (d, $J = 8.5$ Hz, 2H), 5.05 (m, 1H), 4.82 (s, 1H), 4.18 (t, $J = 7.2$ Hz, 2H), 2.74 – 2.52 (m, 8H), 2.44 (t, $J = 7.2$ Hz, 2H), 2.24 (s, 3H), 1.98 – 1.54 (m, 10H), 1.48 – 1.18 (m, 26H), 1.03 (t, $J = 7.2$ Hz, 6H), 0.98 – 0.82 (m, 9H).

31iQ: yellow oil, yield 51%. MS-ESI: calculated for $C_{48}H_{84}N_2O_4$: 753.21, found $[M+H]^+ = 754.02$. 1H NMR (400 MHz, $CDCl_3$) δ 7.99 (d, $J = 8.4$ Hz, 2H), 7.63 (d, $J = 8.4$ Hz, 2H), 5.03 (m, 1H), 4.83 (s, 1H), 4.16 (t, $J = 7.0$ Hz, 2H), 2.73 – 2.51 (m, 8H), 2.42 (t, $J = 7.1$ Hz, 2H), 2.23 (s, 3H), 1.97 – 1.52 (m, 10H), 1.48 – 1.19 (m, 38H), 1.03 (t, $J = 7.1$ Hz, 6H), 0.88 (t, $J = 6.5$ Hz, 9H).

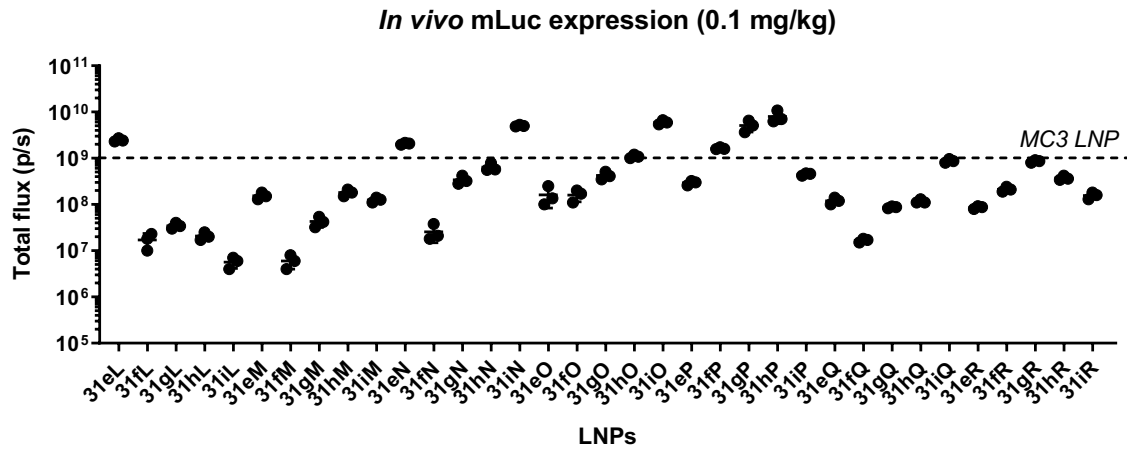
31eR: brown oil, yield 48%. MS-ESI: calculated for $C_{52}H_{92}N_2O_4$: 809.32, found $[M+H]^+ = 809.84$. 1H NMR (400 MHz, $CDCl_3$) δ 7.84 – 7.76 (m, 2H), 7.01 – 6.96 (m, 2H), 4.75 (s, 1H), 4.12 (dt, $J = 12.7, 6.7$ Hz, 2H), 2.81 – 2.48 (m, 8H), 2.44 – 2.16 (m, 5H), 1.87 – 1.51 (m, 12H), 1.47 – 1.17 (m, 42H), 1.07 (t, $J = 7.1$ Hz, 6H), 0.90 (td, $J = 6.0, 3.3$ Hz, 12H).

31fR: brown oil, yield 52%. MS-ESI: calculated for $C_{48}H_{84}N_2O_4$: 753.21, found $[M+H]^+ = 754.12$. 1H NMR (400 MHz, $CDCl_3$) δ 7.98 (d, $J = 8.5$ Hz, 2H), 7.64 (d, $J = 8.5$ Hz, 2H), 4.82 (s, 1H), 4.36 (t, $J = 7.0$ Hz, 2H), 4.12 (t, $J = 6.8$ Hz, 2H), 2.72 – 2.51 (m, 8H), 2.40 (m, 1H), 2.26 (s, 3H), 1.97 – 1.52 (m, 10H), 1.48 – 1.16 (m, 38H), 1.02 (t, $J = 7.2$ Hz, 6H), 0.95 – 0.83 (m, 9H).

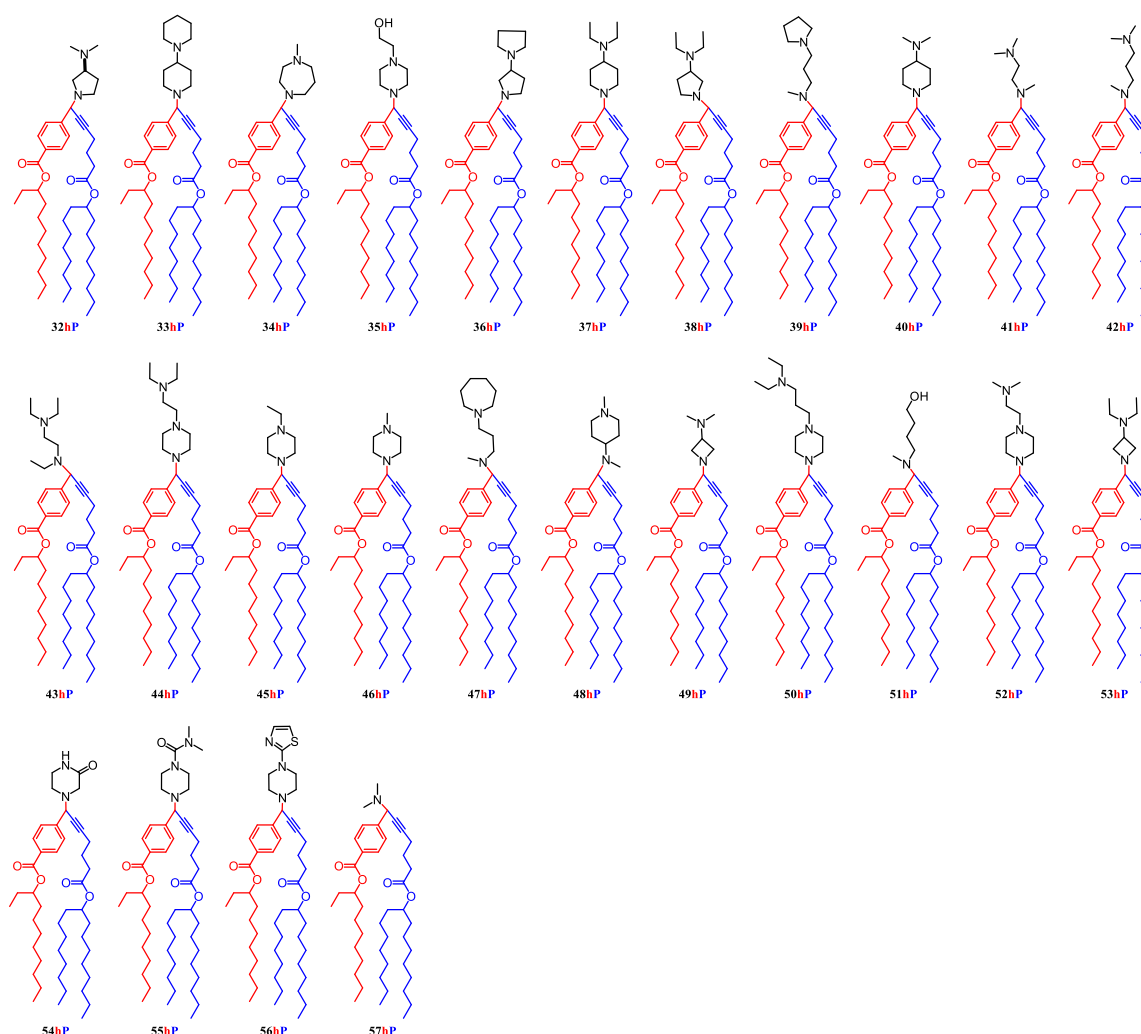
31gR: brown oil, yield 53%. MS-ESI: calculated for $C_{49}H_{86}N_2O_4$: 767.24, found $[M+H]^+ = 768.05$. 1H NMR (400 MHz, $CDCl_3$) δ 7.99 (d, $J = 8.5$ Hz, 2H), 7.66 (d, $J = 8.5$ Hz, 2H), 5.02 (m, 1H), 4.83 (s, 1H), 4.32 (t, $J = 7.1$ Hz, 2H), 2.73 – 2.51 (m, 8H), 2.41 (m, 1H), 2.25 (s, 3H), 1.99 – 1.53 (m, 10H), 1.47 – 1.18 (m, 41H), 1.03 (t, $J = 7.1$ Hz, 6H), 0.94 – 0.82 (m, 9H).

31hR: yellow oil, yield 50%. MS-ESI: calculated for $C_{48}H_{84}N_2O_4$: 753.21, found $[M+H]^+ = 754.02$. 1H NMR (400 MHz, $CDCl_3$) δ 7.99 (d, $J = 8.5$ Hz, 2H), 7.63 (d, $J = 8.5$ Hz, 2H), 5.03 (m, 1H), 4.84 (s, 1H), 4.33 (t, $J = 7.0$ Hz, 2H), 2.73 – 2.52 (m, 8H), 2.42 (m, 1H), 2.25 (s, 3H), 1.98 – 1.51 (m, 12H), 1.48 – 1.16 (m, 34H), 1.03 (t, $J = 7.2$ Hz, 6H), 0.95 – 0.83 (m, 12H).

31iR: brown oil, yield 46%. MS-ESI: calculated for $C_{54}H_{96}N_2O_4$: 837.37, found $[M+H]^+ = 838.24$. 1H NMR (400 MHz, $CDCl_3$) δ 7.98 (d, $J = 8.5$ Hz, 2H), 7.64 (d, $J = 8.4$ Hz, 2H), 5.05 (m, 1H), 4.82 (s, 1H), 4.31 (t, $J = 7.1$ Hz, 2H), 2.73 – 2.51 (m, 8H), 2.44 (m, 1H), 2.26 (s, 3H), 1.97 – 1.52 (m, 12H), 1.49 – 1.15 (m, 46H), 1.02 (t, $J = 7.0$ Hz, 6H), 0.96 – 0.81 (m, 12H).



Supplementary Fig. 8 I *In vivo* mLuc expression. Mice were injected i.v. with mLuc-loaded LNPs at an mRNA dose of 0.1 mg/kg. Images were taken at 4 h post-treatment and total flux was quantified. The dash line indicates the MC3 LNP level. Data are presented as mean $\pm$ SD (n=3 independent biological replicates).



Supplementary Fig. 9 I Chemical structures of A³-lipids in Library 5. These A³-lipids were synthesized based on **Method B**, which were purified and characterized before use.

32hP: brown oil, yield 51%. MS-ESI: calculated for C₄₈H₈₂N₂O₄: 751.19, found [M+H]⁺ = 752.13. ¹H NMR (400 MHz, CDCl₃) δ 8.04 (d, J = 8.3 Hz, 2H), 7.61 (d, J = 7.6 Hz, 2H), 5.13 – 5.05 (m, 1H), 4.91 (dd, J = 8.0, 4.4 Hz, 1H), 4.75 (s, 1H), 2.92 – 2.36 (m, 13H), 2.00 – 1.48 (m, 14H), 1.43 – 1.18 (m, 36H), 0.97 (ddd, J = 9.0, 7.0, 1.6 Hz, 3H), 0.89 (tt, J = 7.1, 2.6 Hz, 9H).

33hP: brown oil, yield 45%. MS-ESI: calculated for C₅₂H₈₈N₂O₄: 805.29, found [M+H]⁺ = 806.08. ¹H NMR (400 MHz, CDCl₃) δ 8.06 – 8.01 (m, 2H), 7.63 (d, J = 8.2 Hz, 2H), 5.09 (p, J = 6.2 Hz, 1H), 4.90 (h, J = 6.7 Hz, 1H), 4.66 (s, 1H), 3.10 – 2.10 (m, 11H), 2.00 – 1.46 (m, 22H), 1.38 – 1.18 (m, 36H), 0.97 (t, J = 7.4 Hz, 3H), 0.89 (td, J = 5.8, 3.5 Hz, 9H).

34hP: brown oil, yield 42%. MS-ESI: calculated for C₄₈H₈₂N₂O₄: 751.19, found [M+H]⁺ = 752.09. ¹H NMR (400 MHz, CDCl₃) δ 8.04 (d, J = 8.4 Hz, 2H), 7.69 (d, J = 8.2 Hz, 2H), 5.09 (p, J = 6.1 Hz, 1H), 4.91 (p, J = 6.3 Hz, 1H), 4.73 (s, 1H), 2.93 – 2.23 (m, 13H), 2.00 – 1.47 (m, 14H), 1.30 – 1.25 (m, 36H), 0.97 (t, J = 7.4 Hz, 3H), 0.93 – 0.85 (m, 9H).

35hP: brown oil, yield 40%. MS-ESI: calculated for C₄₈H₈₂N₂O₅: 767.19, found [M+H]⁺ = 768.09. ¹H NMR (400 MHz, CDCl₃) δ 8.08 – 8.01 (m, 2H), 7.63 (d, J = 8.2 Hz, 2H), 5.10 (q, J = 6.4 Hz, 1H), 4.95 – 4.83 (m, 1H), 4.64 (s, 1H), 3.45 (t, J = 8.0 Hz, 2H), 3.08 – 2.18 (m, 12H), 2.04 – 1.47 (m, 12H), 1.30 – 1.25 (m, 36H), 0.97 (td, J = 7.4, 3.6 Hz, 3H), 0.89 (tq, J = 5.5, 3.3 Hz, 9H).

36hP: brown oil, yield 48%. MS-ESI: calculated for C₅₀H₈₄N₂O₄: 777.23, found [M+H]⁺ = 777.55. ¹H NMR (400 MHz, CDCl₃) δ 8.03 (d, J = 8.2 Hz, 2H), 7.61 (t, J = 8.1 Hz, 2H), 5.13 – 5.05 (m, 1H), 4.90 (t, J = 6.2 Hz,

1H), 4.71 (s, 1H), 2.95 – 2.21 (m, 11H), 2.03 – 1.46 (m, 18H), 1.30 – 1.25 (m, 36H), 1.01 – 0.94 (m, 3H), 0.90 (dp, J = 7.2, 2.6 Hz, 9H).

37hP: brown oil, yield 46%. MS-ESI: calculated for $C_{51}H_{88}N_2O_4$: 793.28, found $[M+H]^+ = 794.25$. 1H NMR (400 MHz, $CDCl_3$) δ 8.04 (d, J = 8.4 Hz, 2H), 7.63 (d, J = 8.2 Hz, 2H), 5.10 (q, J = 6.4 Hz, 1H), 4.89 (t, J = 6.2 Hz, 1H), 4.68 (s, 1H), 2.72 – 2.24 (m, 11H), 2.00 – 1.46 (m, 16H), 1.32 – 1.26 (m, 36H), 1.02 (t, J = 7.2 Hz, 6H), 0.94 (t, J = 7.4 Hz, 3H), 0.87 (td, J = 6.9, 2.4 Hz, 9H).

38hP: brown oil, yield 51%. MS-ESI: calculated for $C_{50}H_{86}N_2O_4$: 779.25, found $[M+H]^+ = 780.21$. 1H NMR (400 MHz, $CDCl_3$) δ 8.03 (d, J = 8.5 Hz, 2H), 7.63 (d, J = 8.4 Hz, 2H), 5.10 (m, 1H), 4.90 (m, 1H), 4.65 (s, 1H), 2.76 – 2.22 (m, 11H), 1.99 – 1.46 (m, 14H), 1.31 – 1.26 (m, 36H), 1.01 (t, J = 7.2 Hz, 6H), 0.95 (t, J = 7.2 Hz, 3H), 0.88 (m, 9H).

39hP: brown oil, yield 53%. MS-ESI: calculated for $C_{50}H_{86}N_2O_4$: 779.25, found $[M+H]^+ = 780.12$. 1H NMR (400 MHz, $CDCl_3$) δ 8.05 (m, 2H), 7.62 (d, J = 8.3 Hz, 2H), 5.09 (q, J = 6.4 Hz, 1H), 4.89 (t, J = 6.2 Hz, 1H), 4.66 (s, 1H), 2.63 – 2.23 (m, 13H), 1.98 – 1.46 (m, 18H), 1.28 (m, 36H), 0.96 (t, J = 7.2 Hz, 3H), 0.88 (td, J = 7.1, 2.4 Hz, 9H).

40hP: brown oil, yield 50%. MS-ESI: calculated for $C_{49}H_{84}N_2O_4$: 765.22, found $[M+H]^+ = 766.14$. 1H NMR (400 MHz, $CDCl_3$) δ 8.04 (d, J = 8.2 Hz, 2H), 7.63 (d, J = 7.6 Hz, 2H), 5.08 (m, 1H), 4.91 (dd, J = 8.0, 4.4 Hz, 1H), 4.73 (s, 1H), 2.62 – 2.22 (m, 13H), 2.01 – 1.45 (m, 16H), 1.29 (m, 36H), 0.97 (t, J = 7.0 Hz, 3H), 0.89 (tt, J = 7.2, 2.6 Hz, 9H).

41hP: brown oil, yield 52%. MS-ESI: calculated for $C_{47}H_{82}N_2O_4$: 739.18, found $[M+H]^+ = 739.98$. 1H NMR (400 MHz, $CDCl_3$) δ 8.04 (d, J = 8.4 Hz, 2H), 7.62 (d, J = 7.8 Hz, 2H), 5.08 (m, 1H), 4.92 (m, 1H), 4.75 (s, 1H), 2.47 – 2.21 (m, 15H), 2.00 – 1.46 (m, 12H), 1.29 (m, 36H), 0.96 (t, J = 7.1 Hz, 3H), 0.90 (tt, J = 7.2, 2.4 Hz, 9H).

42hP: brown oil, yield 51%. MS-ESI: calculated for $C_{48}H_{84}N_2O_4$: 753.21, found $[M+H]^+ = 754.13$. 1H NMR (400 MHz, $CDCl_3$) δ 8.05 (d, J = 8.3 Hz, 2H), 7.63 (d, J = 7.9 Hz, 2H), 5.10 (m, 1H), 4.91 (m, 1H), 4.78 (s, 1H), 2.49 – 2.18 (m, 15H), 1.99 – 1.46 (m, 14H), 1.28 (m, 36H), 0.96 (t, J = 7.2 Hz, 3H), 0.93 – 0.82 (m, 9H).

43hP: brown oil, yield 42%. MS-ESI: calculated for $C_{50}H_{88}N_2O_4$: 781.26, found $[M+H]^+ = 782.15$. 1H NMR (400 MHz, $CDCl_3$) δ 8.04 – 7.96 (m, 2H), 7.65 (d, J = 7.9 Hz, 2H), 5.09 (m, 1H), 4.90 (m, 1H), 4.78 (s, 1H), 2.51 – 2.19 (m, 12H), 2.01 – 1.45 (m, 12H), 1.27 (m, 36H), 1.02 (t, J = 7.2 Hz, 9H), 0.95 (t, J = 7.2 Hz, 3H), 0.92 – 0.81 (m, 9H).

44hP: brown oil, yield 44%. MS-ESI: calculated for $C_{52}H_{91}N_3O_4$: 822.32, found $[M+H]^+ = 823.26$. 1H NMR (400 MHz, $CDCl_3$) δ 8.06 (d, J = 8.4 Hz, 2H), 7.65 (d, J = 7.8 Hz, 2H), 5.08 (m, 1H), 4.90 (m, 1H), 4.76 (s, 1H), 2.64 – 2.23 (m, 18H), 2.00 – 1.46 (m, 12H), 1.29 (m, 36H), 1.01 (t, J = 7.2 Hz, 6H), 0.95 (t, J = 7.2 Hz, 3H), 0.93 – 0.82 (m, 9H).

45hP: brown oil, yield 49%. MS-ESI: calculated for $C_{48}H_{82}N_2O_4$: 751.19, found $[M+H]^+ = 752.14$. 1H NMR (400 MHz, $CDCl_3$) δ 8.05 (d, J = 8.3 Hz, 2H), 7.66 (d, J = 8.0 Hz, 2H), 5.10 (m, 1H), 4.92 (m, 1H), 4.77 (s, 1H), 2.73 – 2.31 (m, 12H), 1.99 – 1.44 (m, 12H), 1.28 (m, 36H), 1.04 (t, J = 7.2 Hz, 3H), 0.96 (t, J = 7.1 Hz, 3H), 0.93 – 0.83 (m, 9H).

46hP: yellow oil, yield 52%. MS-ESI: calculated for $C_{47}H_{80}N_2O_4$: 737.17, found $[M+H]^+ = 738.10$. 1H NMR (400 MHz, $CDCl_3$) δ 8.06 (d, J = 8.2 Hz, 2H), 7.66 (d, J = 7.9 Hz, 2H), 5.08 (m, 1H), 4.92 (m, 1H), 4.76 (s, 1H), 2.72 – 2.20 (m, 13H), 1.97 – 1.44 (m, 12H), 1.28 (m, 36H), 0.94 (t, J = 7.1 Hz, 3H), 0.88 (tt, J = 7.2, 2.5 Hz, 9H).

47hP: brown oil, yield 50%. MS-ESI: calculated for $C_{52}H_{90}N_2O_4$: 807.30, found $[M+H]^+ = 808.17$. 1H NMR (400 MHz, $CDCl_3$) δ 8.06 (d, J = 8.1 Hz, 2H), 7.65 (d, J = 7.9 Hz, 2H), 5.10 (m, 1H), 4.87 (m, 1H), 4.76 (s, 1H), 3.14 – 2.24 (m, 13H), 1.99 – 1.46 (m, 20H), 1.28 (m, 36H), 0.96 (t, J = 7.1 Hz, 3H), 0.92 – 0.81 (m, 9H).

48hP: brown oil, yield 41%. MS-ESI: calculated for $C_{49}H_{84}N_2O_4$: 765.22, found $[M+H]^+ = 766.13$. 1H NMR (400 MHz, $CDCl_3$) δ 8.08 – 7.99 (m, 2H), 7.65 (d, J = 8.1 Hz, 2H), 5.06 (m, 1H), 4.89 (m, 1H), 4.78 (s, 1H), 2.68 – 2.22 (m, 13H), 1.98 – 1.42 (m, 16H), 1.29 (m, 36H), 0.96 (t, J = 7.1 Hz, 3H), 0.92 – 0.82 (m, 9H).

49hP: brown oil, yield 51%. MS-ESI: calculated for $C_{47}H_{80}N_2O_4$: 737.17, found $[M+H]^+ = 728.23$. 1H NMR (400 MHz, $CDCl_3$) δ 8.05 (d, $J = 8.4$ Hz, 2H), 7.64 (d, $J = 8.2$ Hz, 2H), 5.11 (q, $J = 6.4$ Hz, 1H), 4.91 (t, $J = 6.3$ Hz, 1H), 4.69 (s, 1H), 3.58 – 3.21 (m, 5H), 2.28 (t, $J = 7.0$ Hz, 2H), 2.23 (s, 6H), 1.98 – 1.46 (m, 12H), 1.32 – 1.24 (m, 36H), 0.96 (t, $J = 7.3$ Hz, 3H), 0.88 (td, $J = 6.9, 2.4$ Hz, 9H).

50hP: brown oil, yield 47%. MS-ESI: calculated for $C_{53}H_{93}N_3O_4$: 836.34, found $[M+H]^+ = 837.25$. 1H NMR (400 MHz, $CDCl_3$) δ 8.05 (d, $J = 8.3$ Hz, 2H), 7.66 (d, $J = 8.0$ Hz, 2H), 5.08 (m, 1H), 4.89 (m, 1H), 4.78 (s, 1H), 3.04 – 2.28 (m, 18H), 1.98 – 1.45 (m, 14H), 1.29 (m, 36H), 1.07 (t, $J = 7.2$ Hz, 6H), 0.95 (t, $J = 7.2$ Hz, 3H), 0.84 (m, 9H).

51hP: brown oil, yield 40%. MS-ESI: calculated for $C_{47}H_{81}NO_5$: 740.17, found $[M+H]^+ = 741.18$. 1H NMR (400 MHz, $CDCl_3$) δ 8.07 (m, 2H), 7.65 (d, $J = 8.3$ Hz, 2H), 5.09 (m, 1H), 4.92 (m, 1H), 4.80 (s, 1H), 3.64 (t, $J = 7.2$ Hz, 2H), 2.48 (m, 7H), 2.00 – 1.46 (m, 16H), 1.29 (m, 36H), 0.97 (t, $J = 7.4$ Hz, 3H), 0.93 – 0.84 (m, 9H).

52hP: brown oil, yield 48%. MS-ESI: calculated for $C_{50}H_{87}N_3O_4$: 794.26, found $[M+H]^+ = 795.17$. 1H NMR (400 MHz, $CDCl_3$) δ 8.05 (d, $J = 8.4$ Hz, 2H), 7.66 (d, $J = 8.0$ Hz, 2H), 5.07 (m, 1H), 4.88 (m, 1H), 4.76 (s, 1H), 2.71 – 2.24 (m, 20H), 1.98 – 1.42 (m, 12H), 1.29 (m, 36H), 0.96 (t, $J = 7.1$ Hz, 3H), 0.92 – 0.82 (m, 9H).

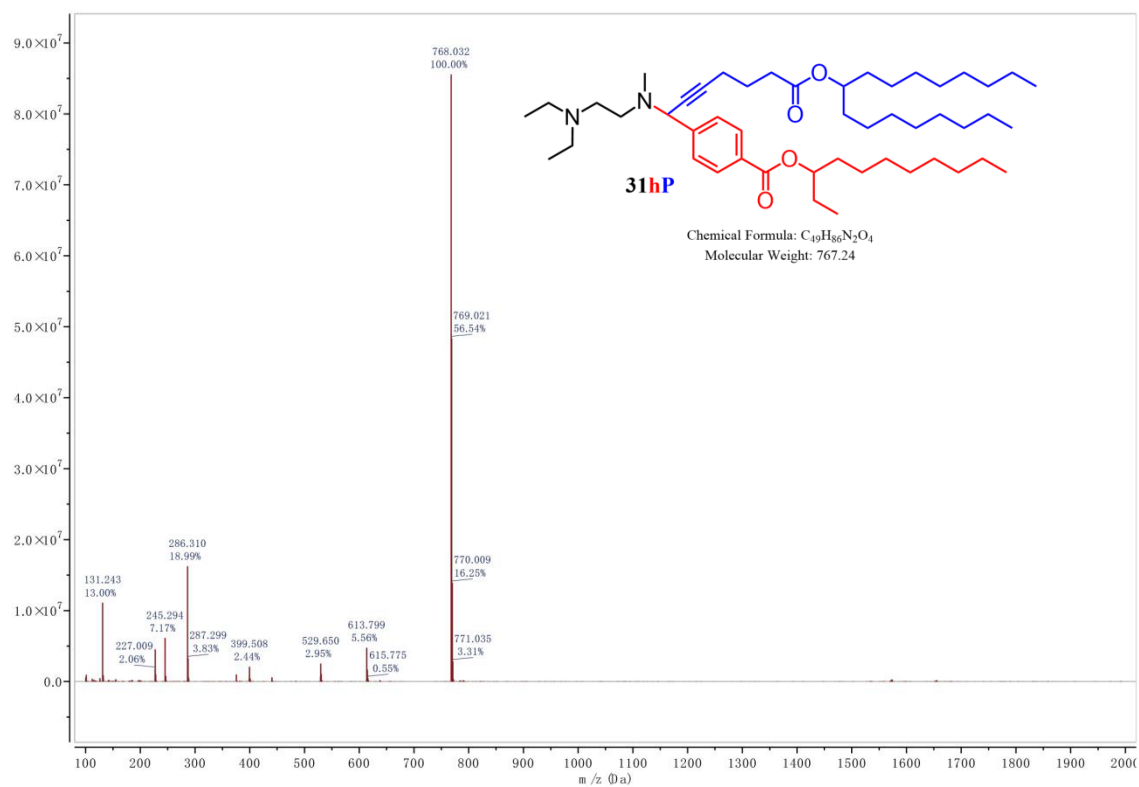
53hP: brown oil, yield 47%. MS-ESI: calculated for $C_{49}H_{84}N_2O_4$: 765.22, found $[M+H]^+ = 766.14$. 1H NMR (400 MHz, $CDCl_3$) δ 8.04 (d, $J = 8.3$ Hz, 2H), 7.65 (d, $J = 7.9$ Hz, 2H), 5.08 (m, 1H), 4.87 (m, 1H), 4.76 (s, 1H), 3.54 – 3.21 (m, 5H), 2.69 – 2.31 (m, 6H), 1.99 – 1.44 (m, 12H), 1.28 (m, 36H), 1.04 (t, $J = 7.2$ Hz, 6H), 0.95 (t, $J = 7.4$ Hz, 3H), 0.86 (m, 9H).

54hP: yellow oil, yield 39%. MS-ESI: calculated for $C_{46}H_{76}N_2O_5$: 737.12, found $[M+H]^+ = 737.77$. 1H NMR (400 MHz, $CDCl_3$) δ 8.03 (d, $J = 8.3$ Hz, 2H), 7.55 – 7.50 (m, 2H), 5.07 (m, 1H), 4.90 (t, $J = 6.2$ Hz, 1H), 4.73 (s, 1H), 3.70 – 3.20 (m, 4H), 2.86 (t, $J = 7.2$ Hz, 2H), 2.46 (t, $J = 7.4$ Hz, 2H), 2.01 – 1.48 (m, 12H), 1.30 – 1.25 (m, 36H), 0.97 (m, 3H), 0.89 (td, $J = 6.1, 2.5$ Hz, 9H).

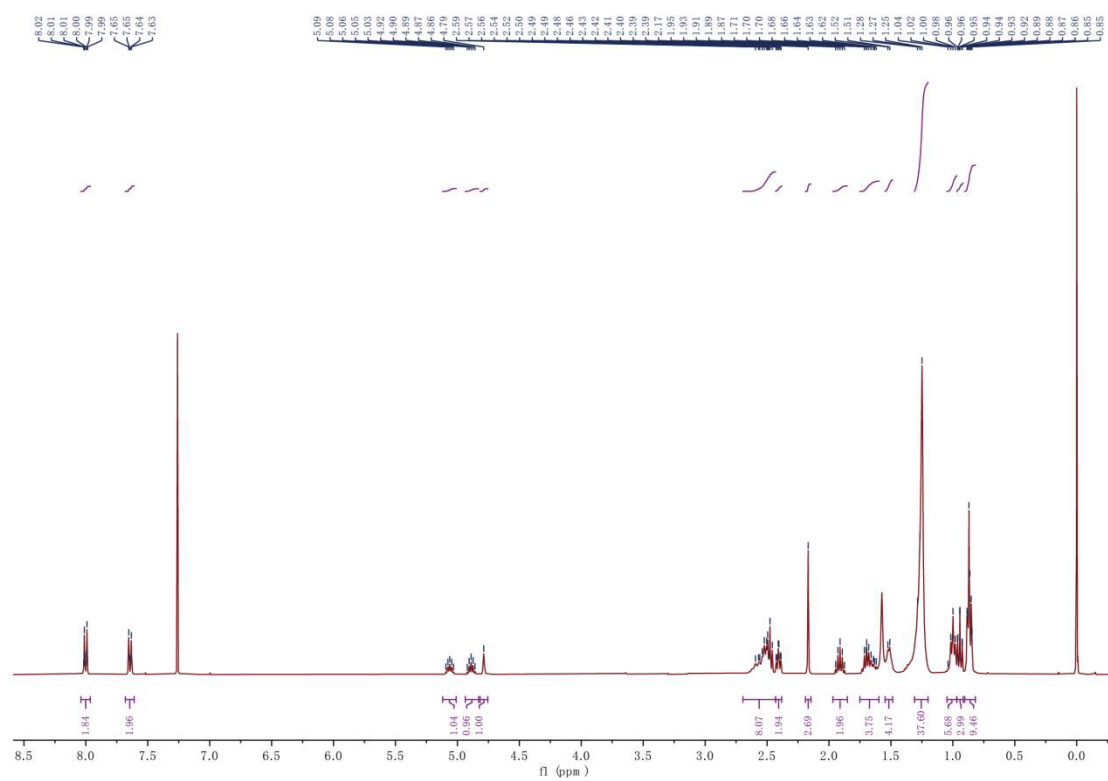
55hP: brown oil, yield 42%. MS-ESI: calculated for $C_{49}H_{83}N_3O_5$: 794.22, found $[M+H]^+ = 795.15$. 1H NMR (400 MHz, $CDCl_3$) δ 8.05 (d, $J = 8.3$ Hz, 2H), 7.55 – 7.50 (m, 2H), 5.08 (q, $J = 6.3$ Hz, 1H), 4.89 (t, $J = 6.1$ Hz, 1H), 4.75 (s, 1H), 3.31 – 2.83 (m, 14H), 2.37 (t, $J = 7.2$ Hz, 2H), 1.99 – 1.46 (m, 12H), 1.29 – 1.24 (m, 36H), 0.96 (t, $J = 7.4$ Hz, 3H), 0.87 (m, 9H).

56hP: dark brown oil, yield 37%. MS-ESI: calculated for $C_{49}H_{79}N_3O_4S$: 806.25, found $[M+H]^+ = 807.17$. 1H NMR (400 MHz, $CDCl_3$) δ 8.06 (d, $J = 8.4$ Hz, 2H), 7.69 (d, $J = 7.8$ Hz, 1H), 7.64 (d, $J = 8.0$ Hz, 2H), 6.84 (d, $J = 7.8$ Hz, 1H), 5.06 (q, $J = 6.4$ Hz, 1H), 4.87 (q, $J = 6.3$ Hz, 1H), 4.74 (s, 1H), 2.88 (s, 8H), 2.38 (t, $J = 7.0$ Hz, 2H), 1.98 – 1.45 (m, 12H), 1.27 (m, 36H), 0.94 (t, $J = 7.2$ Hz, 3H), 0.85 (m, 9H).

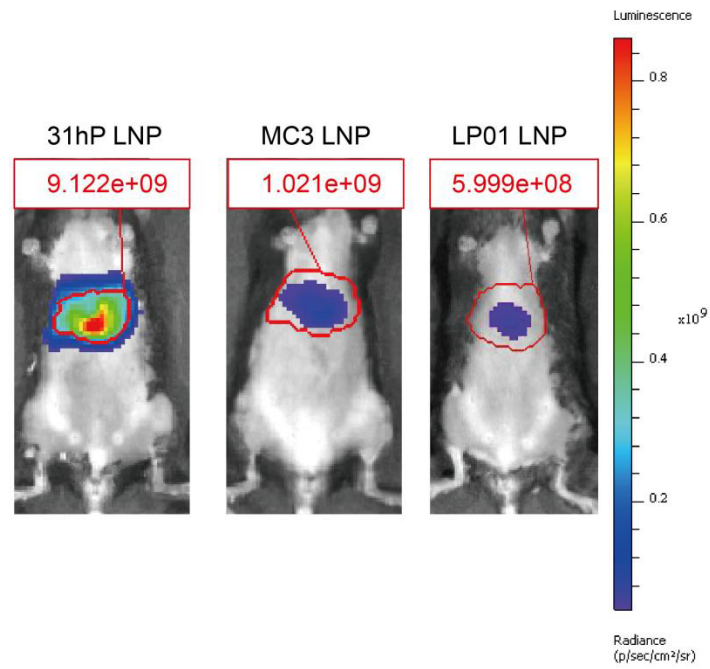
57hP: light yellow oil, yield 42%. MS-ESI: calculated for $C_{44}H_{75}NO_4$: 682.09, found $[M+H]^+ = 682.93$. 1H NMR (400 MHz, $CDCl_3$) δ 8.03 (d, $J = 8.2$ Hz, 2H), 7.65 (d, $J = 8.0$ Hz, 2H), 5.08 (m, 1H), 4.89 (t, $J = 6.3$ Hz, 1H), 4.74 (s, 1H), 2.39 (t, $J = 7.2$ Hz, 2H), 2.28 (s, 6H), 2.00 – 1.46 (m, 12H), 1.29 (m, 36H), 0.95 (t, $J = 7.0$ Hz, 3H), 0.87 (m, 9H).



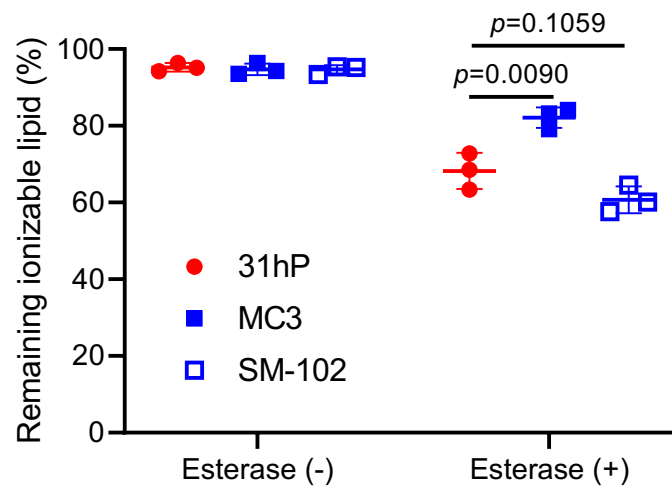
Supplementary Fig. 10 | Mass spectrum of 31hP A³-lipid.



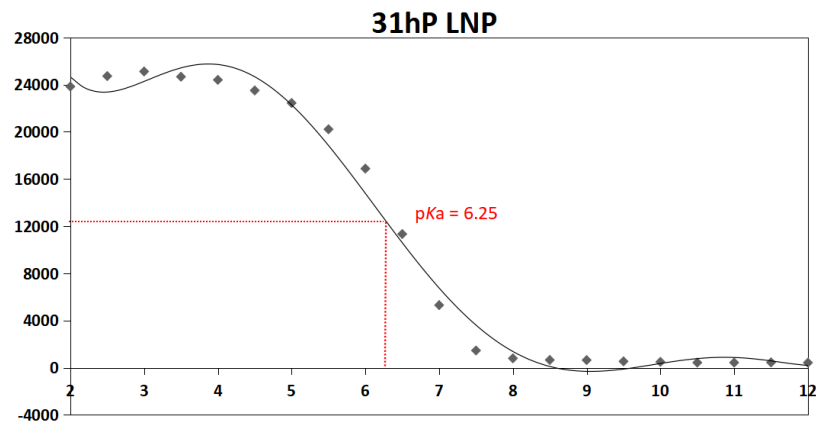
Supplementary Fig. 11 | ^1H NMR spectrum of 31hP A³-lipid in CDCl_3 .



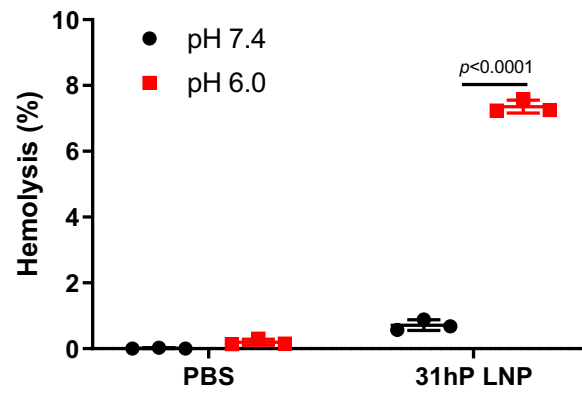
Supplementary Fig. 12 | Representative *in vivo* bioluminescence imaging results of 31 hP LNP, MC3 LNP and LP01 LNP. Mice were i.v. injected with mLuc-loaded LNPs at an mRNA dose of 0.1 mg/kg. Images were taken at 4 h post-treatment. The signal mainly localized in the upper abdomen with predominant liver transfection.



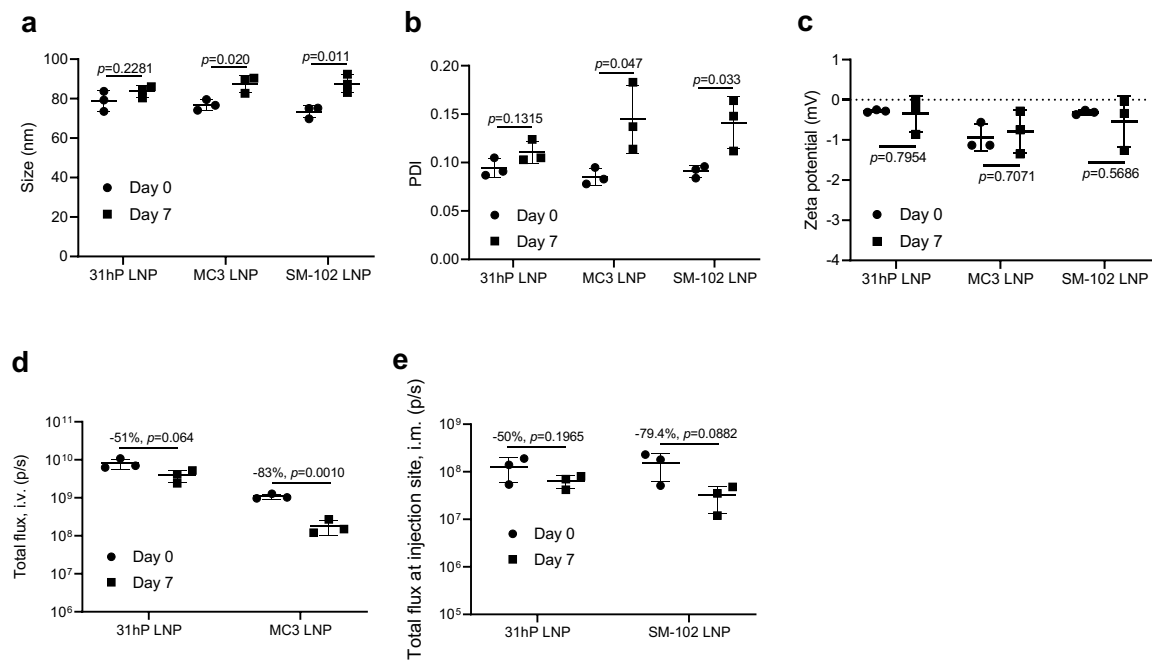
Supplementary Fig. 13 | Esterase-mediated lipid degradation. The remaining ionizable lipid was examined at 12 h post-treatment with or without esterase. Data are presented as mean $\pm$ SD ($n=3$ independent biological replicates). Statistical significance was evaluated by a one-way ANOVA with Tukey's correction.



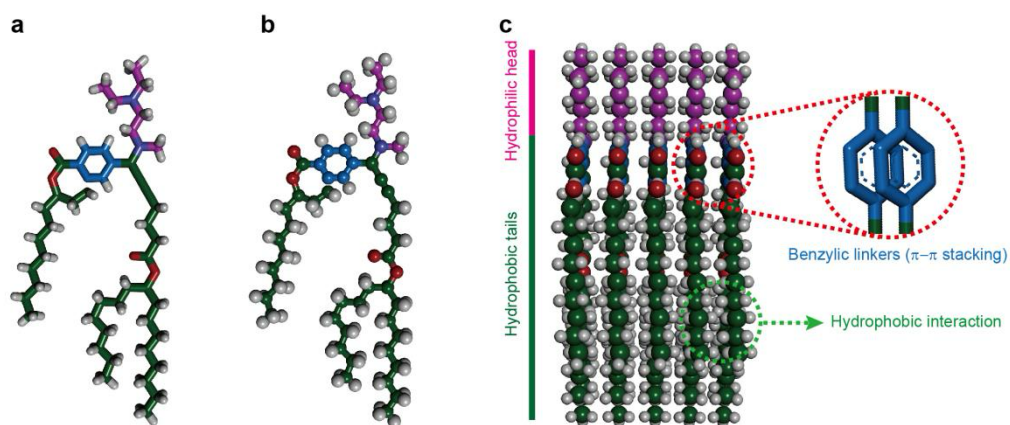
Supplementary Fig. 14 | TNS assay was used to determine the apparent pK_a of 31hP LNP. TNS fluorescence signal corresponds to ionization. pK_a is calculated as the pH corresponding to half of the maximum TNS fluorescence value.



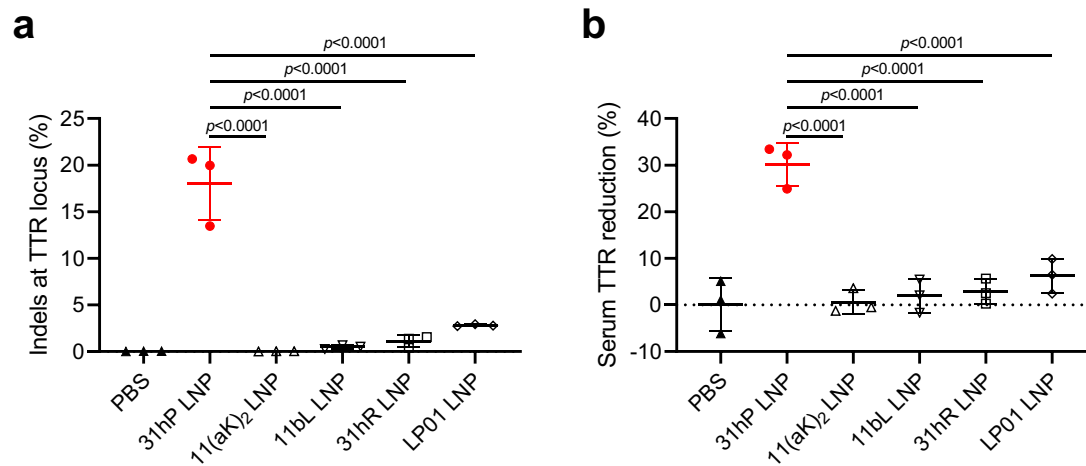
Supplementary Fig. 15 I Hemolysis of 31hP LNP at pH 7.4 or 6.0. RBCs were incubated with 31hP LNP at an mRNA concentration of 3 $\mu\text{g/mL}$ at 37 $^{\circ}\text{C}$ for 1 h. Positive and negative controls were carried out with 0.1% Triton-X (100% hemolysis) and 1 $\times$ PBS (0% hemolysis), respectively. Data are presented as mean $\pm$ SD (n=3 independent biological replicates). Statistical significance was evaluated by a one-way ANOVA with Tukey's correction.



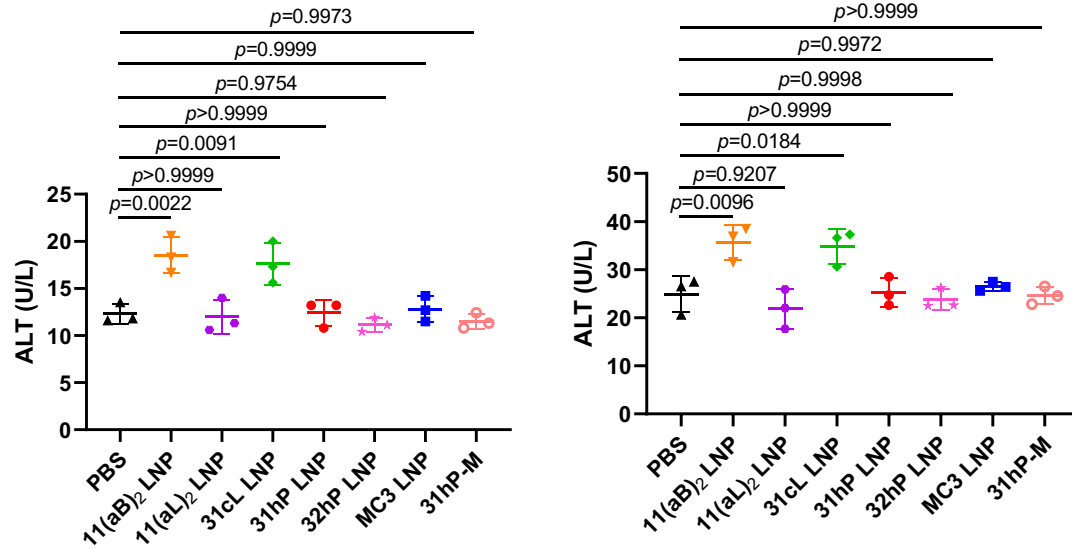
Supplementary Fig. 16 | Stability and durability of LNPs after storage of LNPs at 4 °C for 7 days. a, Hydrodynamic size. **b,** PDI. **c,** Zeta potential. **d,** Total flux after i.v. delivery of mLuc-loaded LNPs. Mice were i.v. injected with mLuc-loaded 31hP LNP or MC3 LNP at an mRNA dose of 0.1 mg/kg. *In vivo* luminescence imaging was performed at 4 h post-treatment and whole-body total flux was quantified. **e,** Total flux at injection site after i.m. delivery of mLuc-loaded LNPs. Mice were i.m. injected with mLuc-loaded 31hP LNP or SM-102 LNP at an mRNA dose of 2 μ g mRNA/mouse. *In vivo* luminescence imaging was performed at 4 h post-treatment and total flux at injection site was quantified. Data are presented as mean $\pm$ SD (n=3 independent biological replicates). Statistical significance was evaluated by a one-way ANOVA with Tukey's correction.



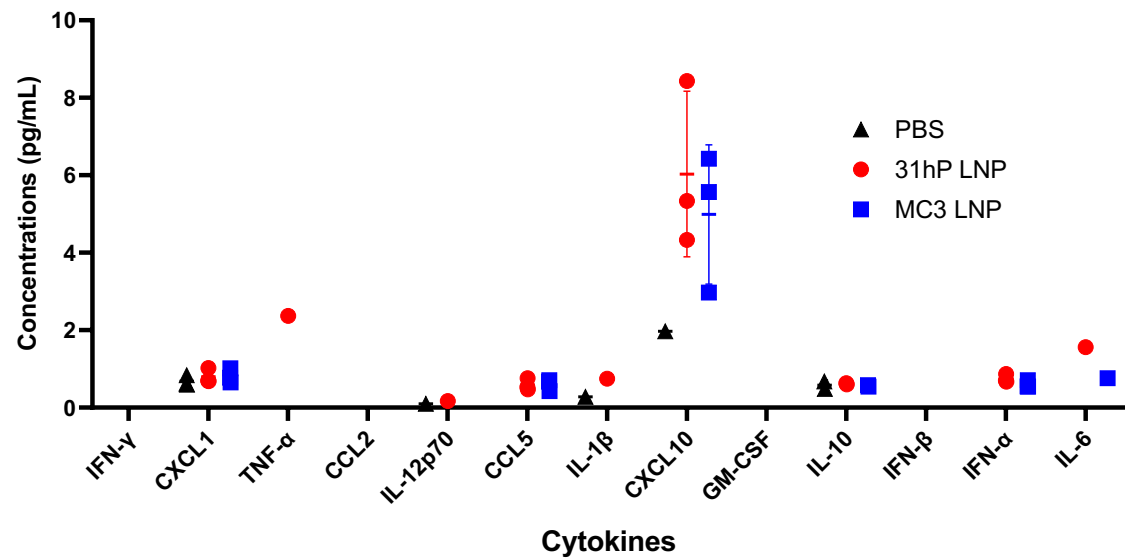
Supplementary Fig. 17 | Molecular modeling of 31hP. **a**, Molecular structure of 31hP in a stick model. **b**, Molecular structure of 31hP in a ball-and-stick model. **c**, Molecular assembly of 31hP driven by hydrophobic interaction and π - π stacking. C in blue, green or pink; O in red; H in white; N in purple.



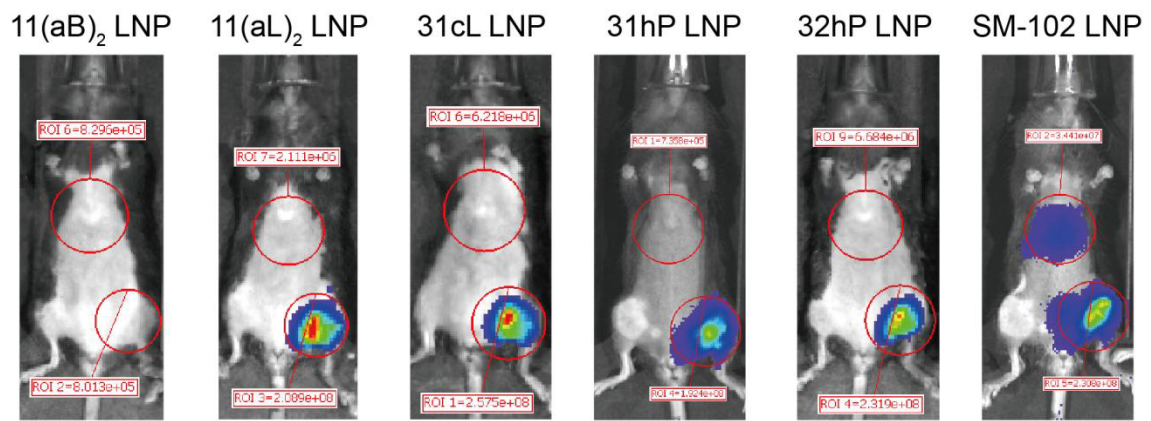
Supplementary Fig. 18 | LNP-enabled TTR gene editing. a, Indels at TTR locus. **b**, Serum TTR. Mice were i.v. injected with LNPs encapsulating Cas9 mRNA/TTR sgRNA (4:1, wt:wt) at a total RNA dose of 1 mg/kg. On day 7, DNA was extracted from the liver to determine on-target indel frequency by next-generation sequencing (NGS), and serum was collected to determine TTR concentration by ELISA. Data are presented as mean $\pm$ SD (n=3 independent biological replicates). Statistical significance was evaluated by a one-way ANOVA with Tukey's correction.



Supplementary Fig. 19 | ALT and AST analysis. Cas9 mRNA/TTR sgRNA-loaded LNPs (1 mg/kg) or 31hP-M (a degradation metabolite of 31hP, 5 mg/kg) were i.v. injected into mice. Serum was collected at 24 h post-treatment for ALT and AST analysis. Data are presented as mean $\pm$ SD (n=3 independent biological replicates). Statistical significance was evaluated by a one-way ANOVA with Tukey's correction.

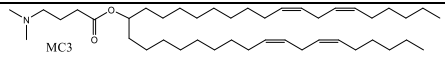
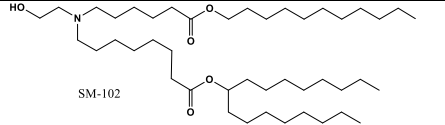
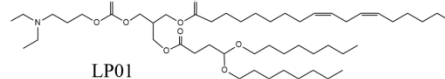


Supplementary Fig. 20 | Cytokine responses. Cas9 mRNA/TTR sgRNA-loaded LNPs (1 mg/kg) were i.v. injected into mice. Serum was collected at 24 h post-treatment for flow-based multiplex cytokine analysis. Cytokines were all at low levels (< 10 pg/mL) after LNP treatment. Data are presented as mean $\pm$ SD (n=3 independent biological replicates). Data point with cytokine level lower than the minimum detectable concentration is not shown on the chart.



Supplementary Fig. 21 | Representative *in vivo* bioluminescence imaging after i.m. injection. mLuc-loaded LNPs (2 μ g mRNA in 50 μ L PBS) were injected into the gastrocnemius muscle. Images were taken at 4 h post-treatment and total flux at injection site or liver region was quantified.

Supplementary Table 1 | Physiochemical parameters of LNPs formulated by microfluidic mixing.

LNP	Ionizable lipid	Size (nm)	PDI	Zeta potential (mV)	EE (%)
11(aB) ₂ LNP	11(aB) ₂	118.6 ± 7.3	0.139	-3.37 ± 1.11	31.4 ± 2.8
11(aL) ₂ LNP	11(aL) ₂	80.3 ± 2.8	0.104	-0.84 ± 0.02	93.2 ± 1.3
31cL LNP	31cL	77.7 ± 1.6	0.128	-1.56 ± 0.45	92.2 ± 0.4
32 hP LNP	32 hP	81.9 ± 5.1	0.098	-2.13 ± 0.27	94.5 ± 1.6
11(aK) ₂ LNP	11(aK) ₂	91.2 ± 4.3	0.124	0.27 ± 0.04	91.6 ± 0.5
11bL LNP	11bL	82.7 ± 3.7	0.085	-2.18 ± 0.75	93.2 ± 0.8
31 hR LNP	31 hR	78.6 ± 2.1	0.107	-1.02 ± 0.01	91.8 ± 1.1
MC3 LNP		76.4 ± 5.2	0.087	-1.01 ± 0.06	92.9 ± 1.1
SM-102 LNP		73.5 ± 4.7	0.092	-0.32 ± 0.02	93.1 ± 1.6
LP01 LNP		69.6 ± 1.1	0.117	-1.24 ± 0.07	95.2 ± 0.8

The chemical structures of MC3, SM-102 and LP01 were shown. The hydrodynamic size, PDI and zeta potential of LNPs were obtained by dynamic light scattering (DLS) measurement in PBS (pH 7.4). The mRNA encapsulation efficiency (EE) was determined using a modified Quant-iT RiboGreen RNA assay. Data are presented as mean ± SD (n=3 independent biological replicates).